

SUPER * TREE

LEARNERS:

a new metalearner for
local data science

LEVI JOHN WOLF `levi.john.wolf@bristol.ac.uk`

WHY DO WE NEED THIS?

GEOGRAPHICAL CLUSTER-REG

UNDERSTANDING QUADTREES

APPLYING QUADTREE REGRESSION

SUPER*TREE LEARNERS

WHY DO WE NEED THIS?

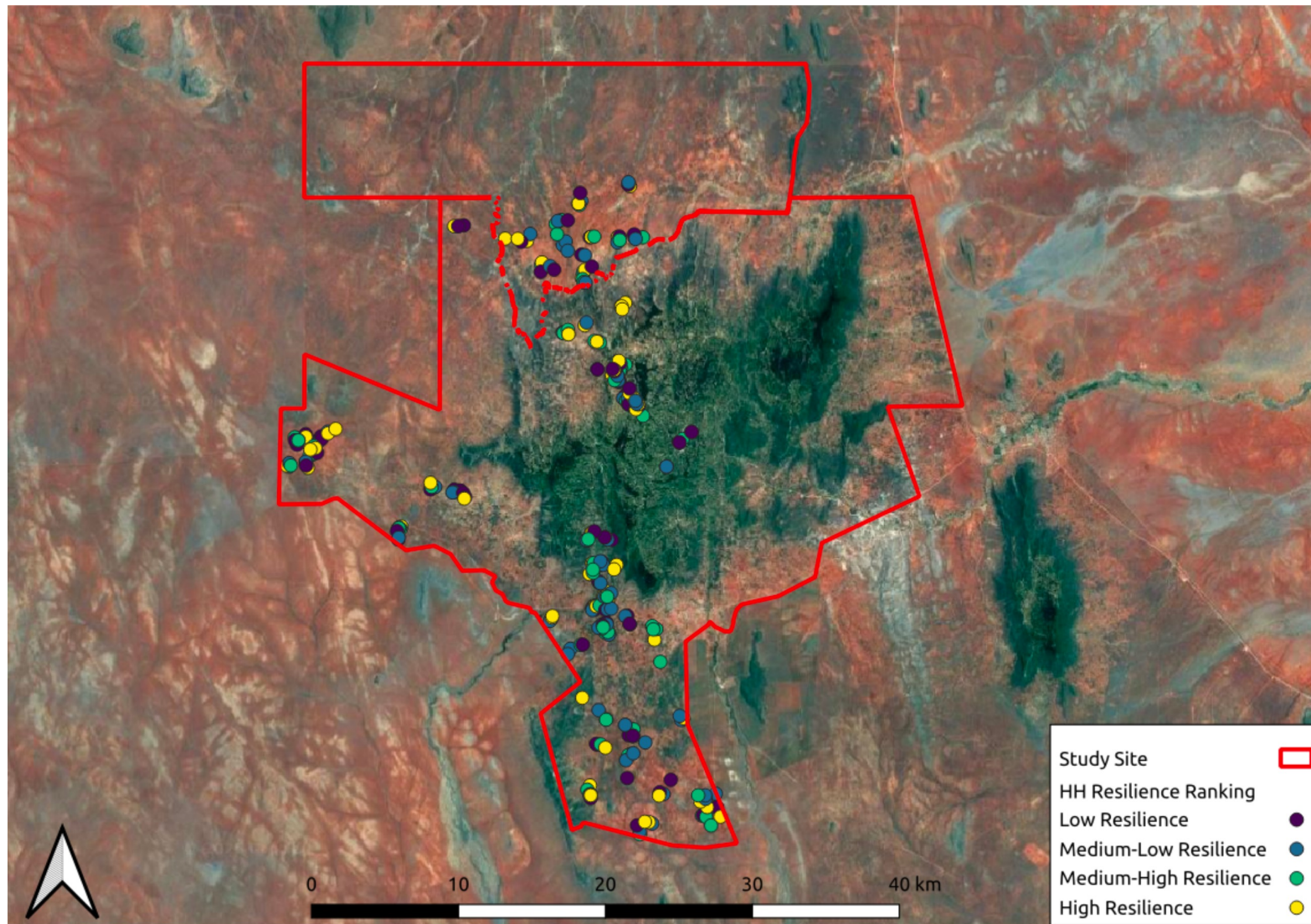
place is not always so obvious

GEOGRAPHICAL CLUSTER-REG

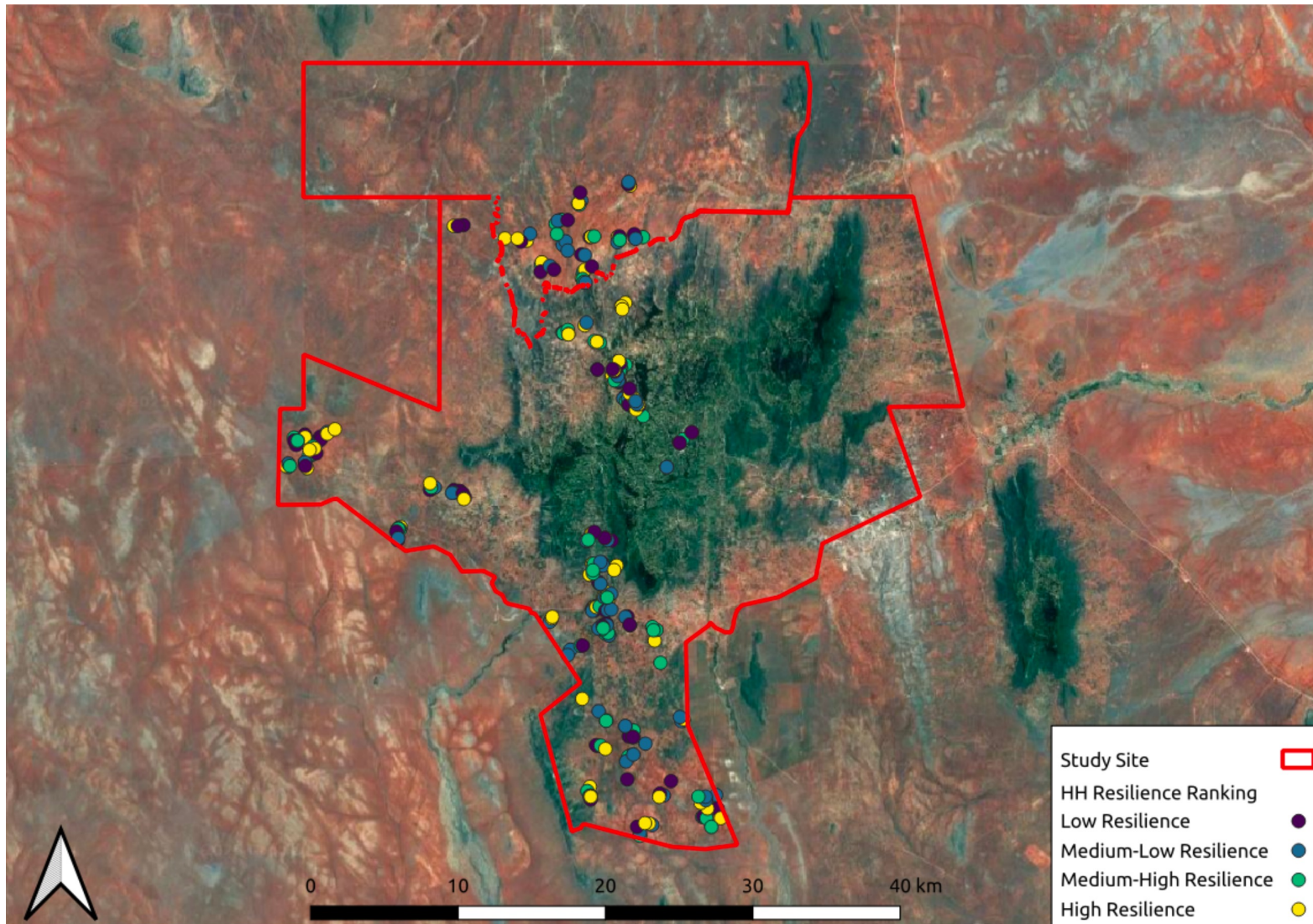
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APPLYING QUADTREE REGRESSION

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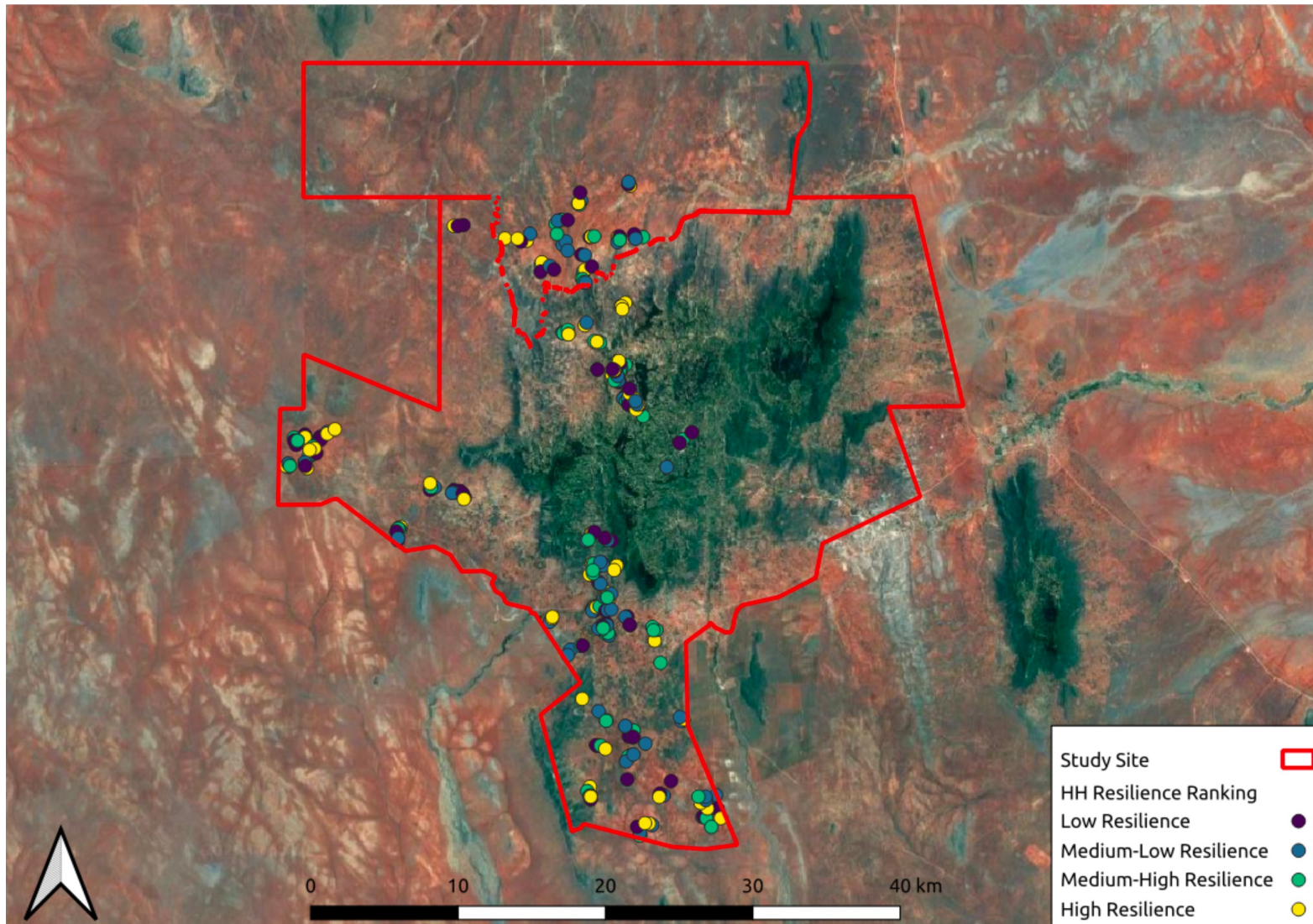
WHAT IS CLUSTERING REGRESSION



Different clusters differ:

- What is “food”?
- Do we farm or herd?
- How do we do that?
- Is water held in common?
- Is there a community safety net?

WHAT IS CLUSTERING REGRESSION



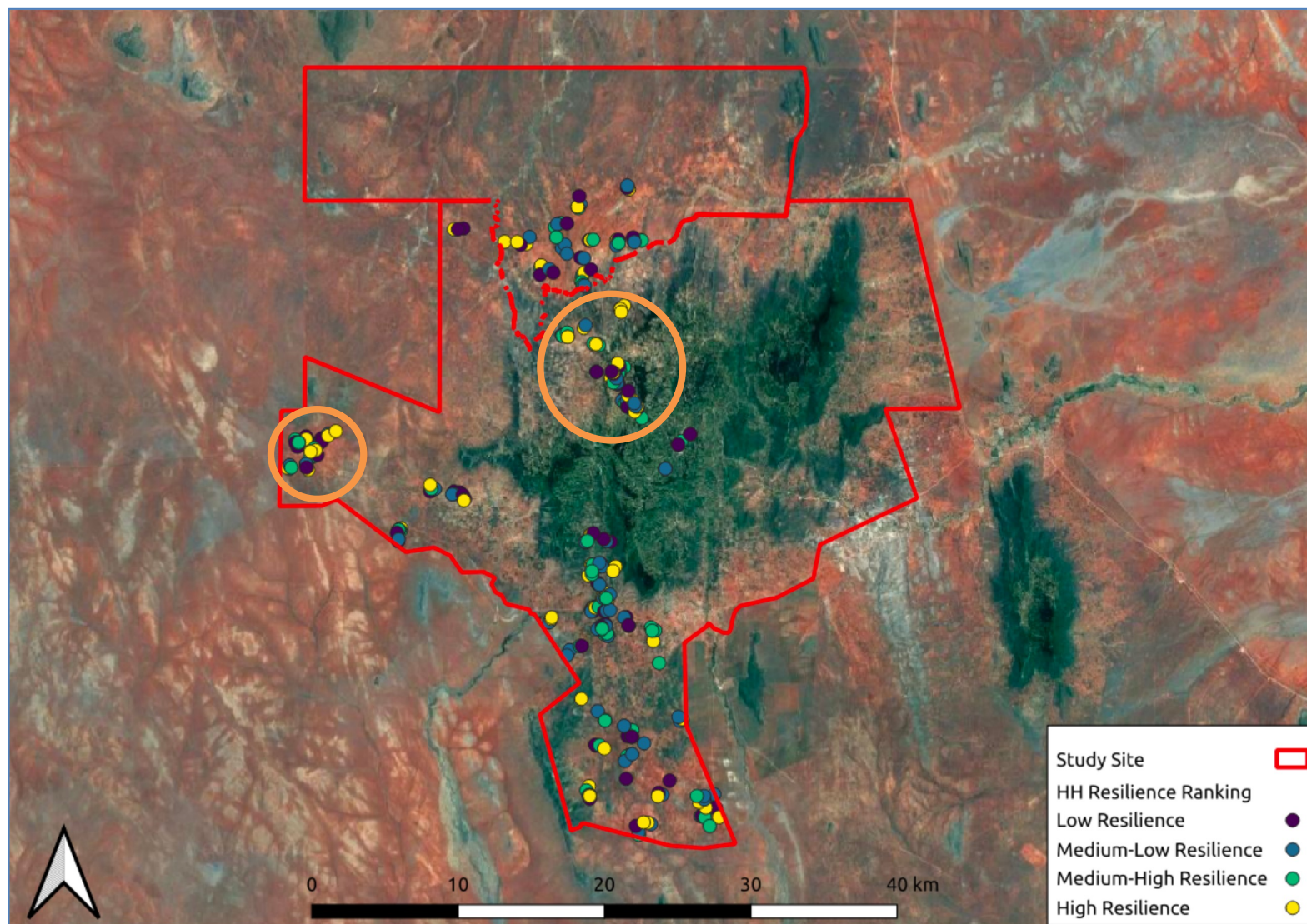
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Nearby households are more similar ...

**Geographically
Weighted
Regression**

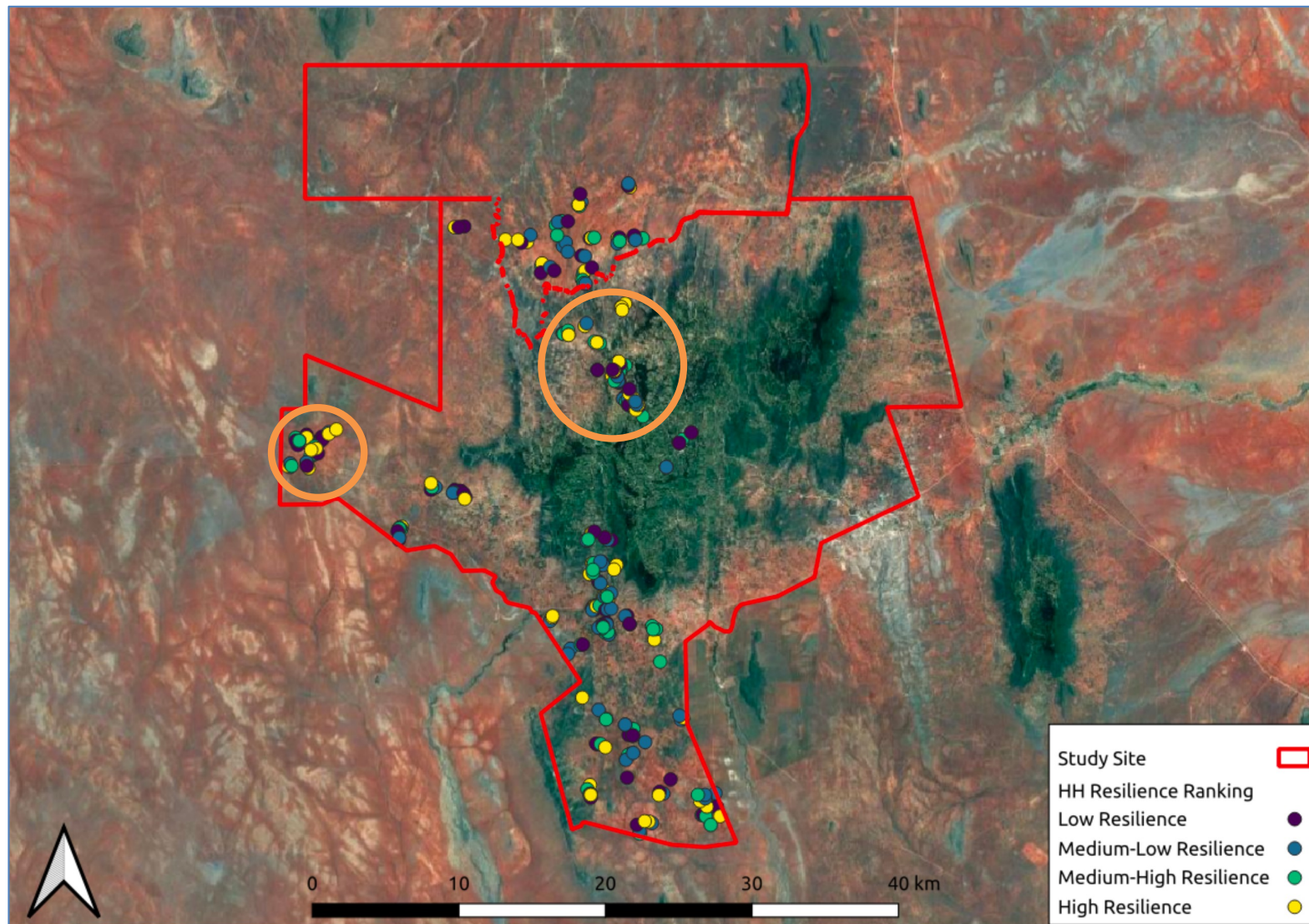
WHAT IS CLUSTERING REGRESSION



Nearby households are more similar ...

Geographically Weighted Regression

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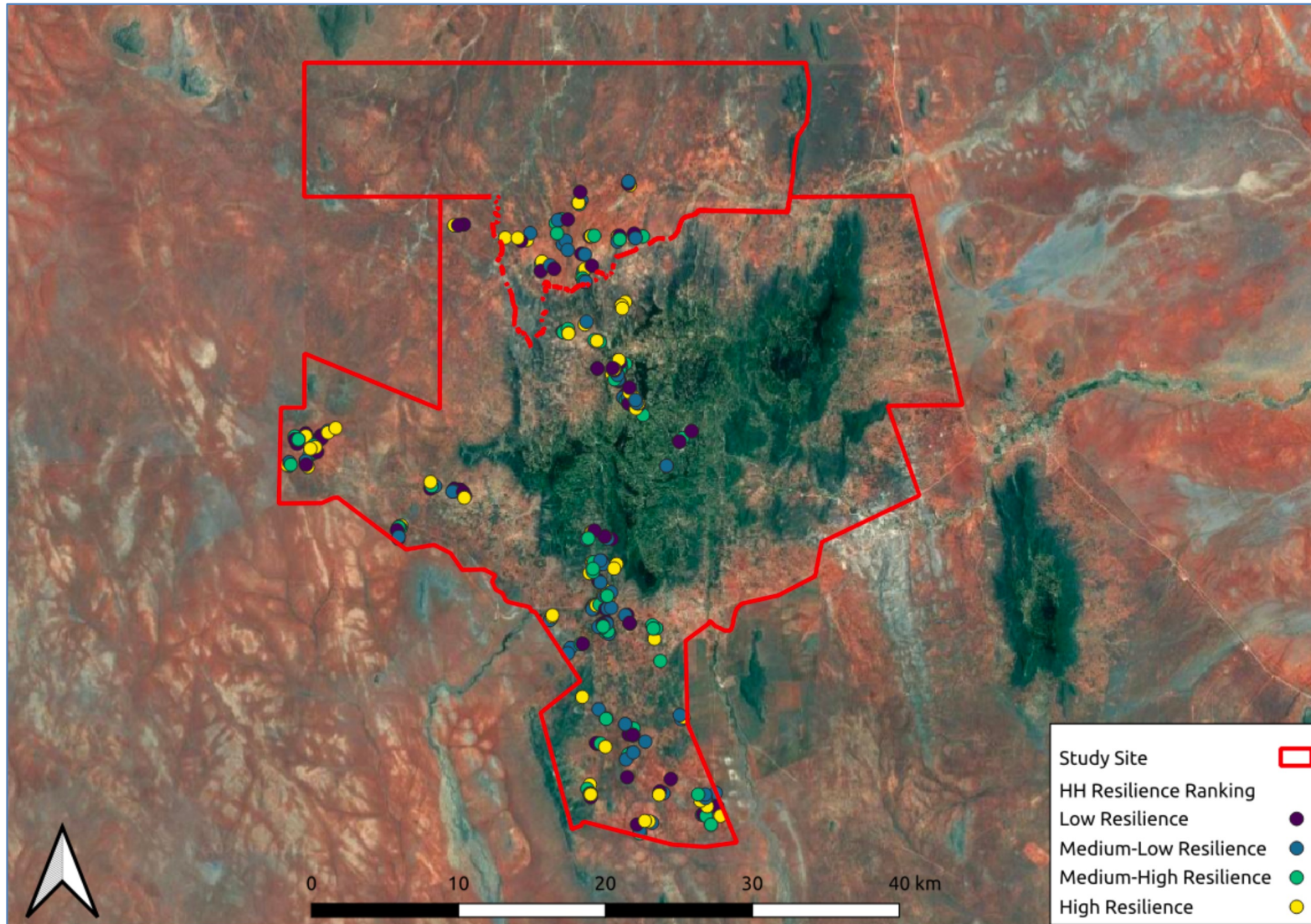


Often,
separation
is poor!

Nearby
households are
more similar ...

**Geographically
Weighted
Regression**

WHAT IS CLUSTERING REGRESSION

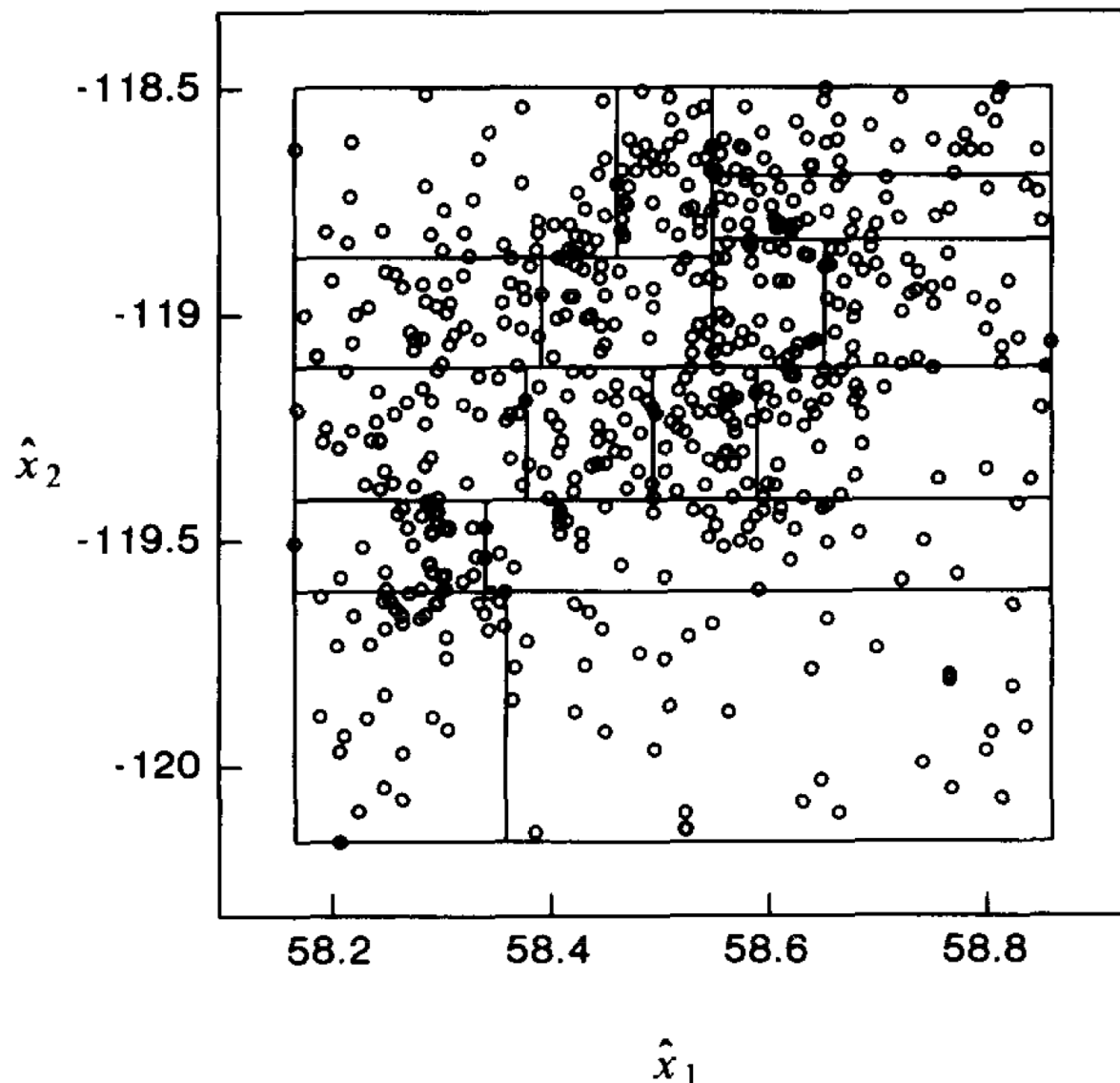


Instead, let's
find who's
similar in
 $y \sim X$ and
group'em!

Nearby
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WHAT IS CLUSTERING REGRESSION



**Instead, let's
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REGRESSION BY LOCAL FITTING
Methods, Properties, and Computational Algorithms

William S. CLEVELAND

AT&T Bell Laboratories, Murray Hill, NJ 07974, USA

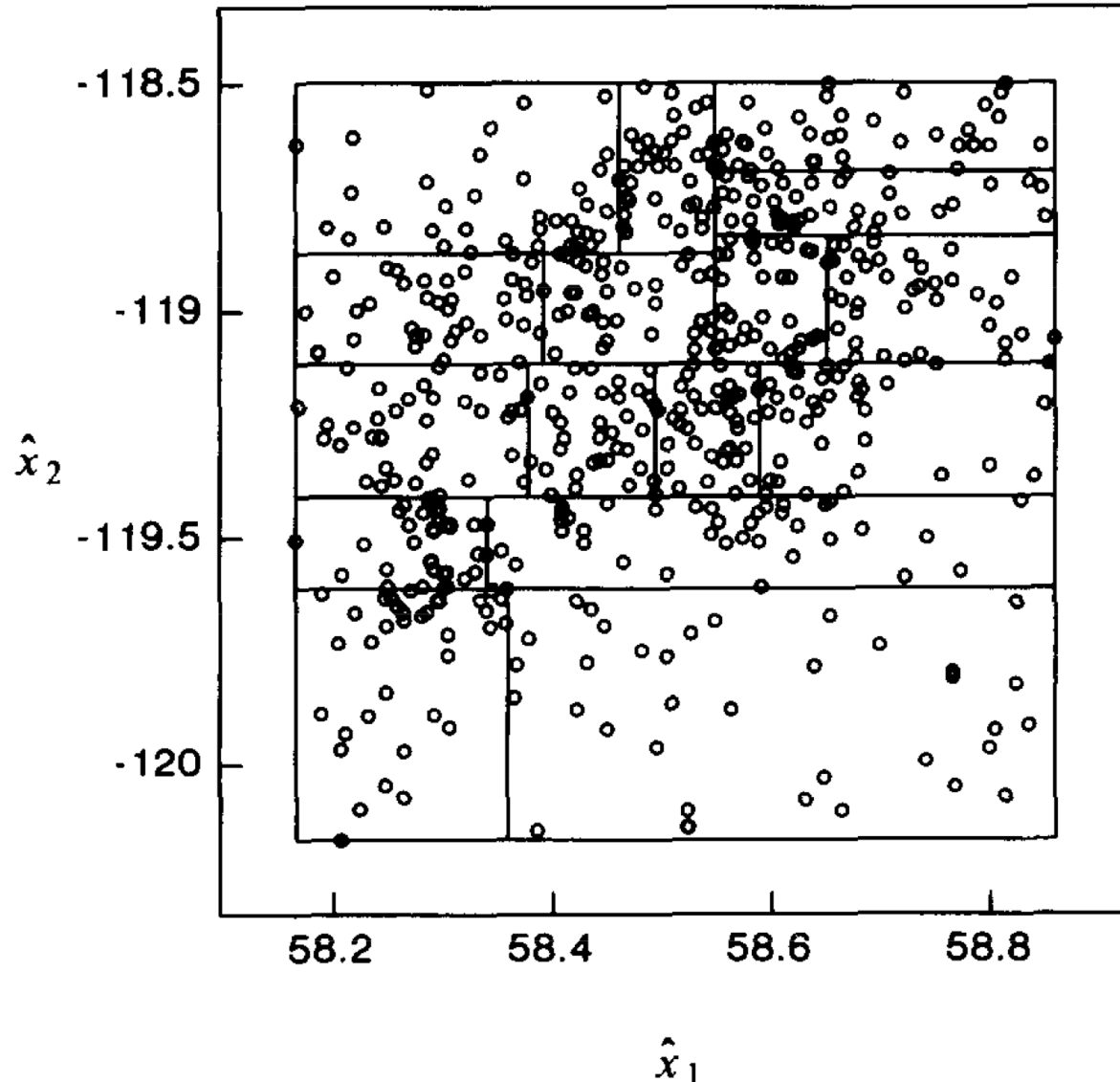
Susan J. DEVLIN

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WHAT IS CLUSTERING REGRESSION



LOWESS Idea:

Fit a spatial tree on feature data
Estimate separately for each leaf
Smooth between nearby leaves

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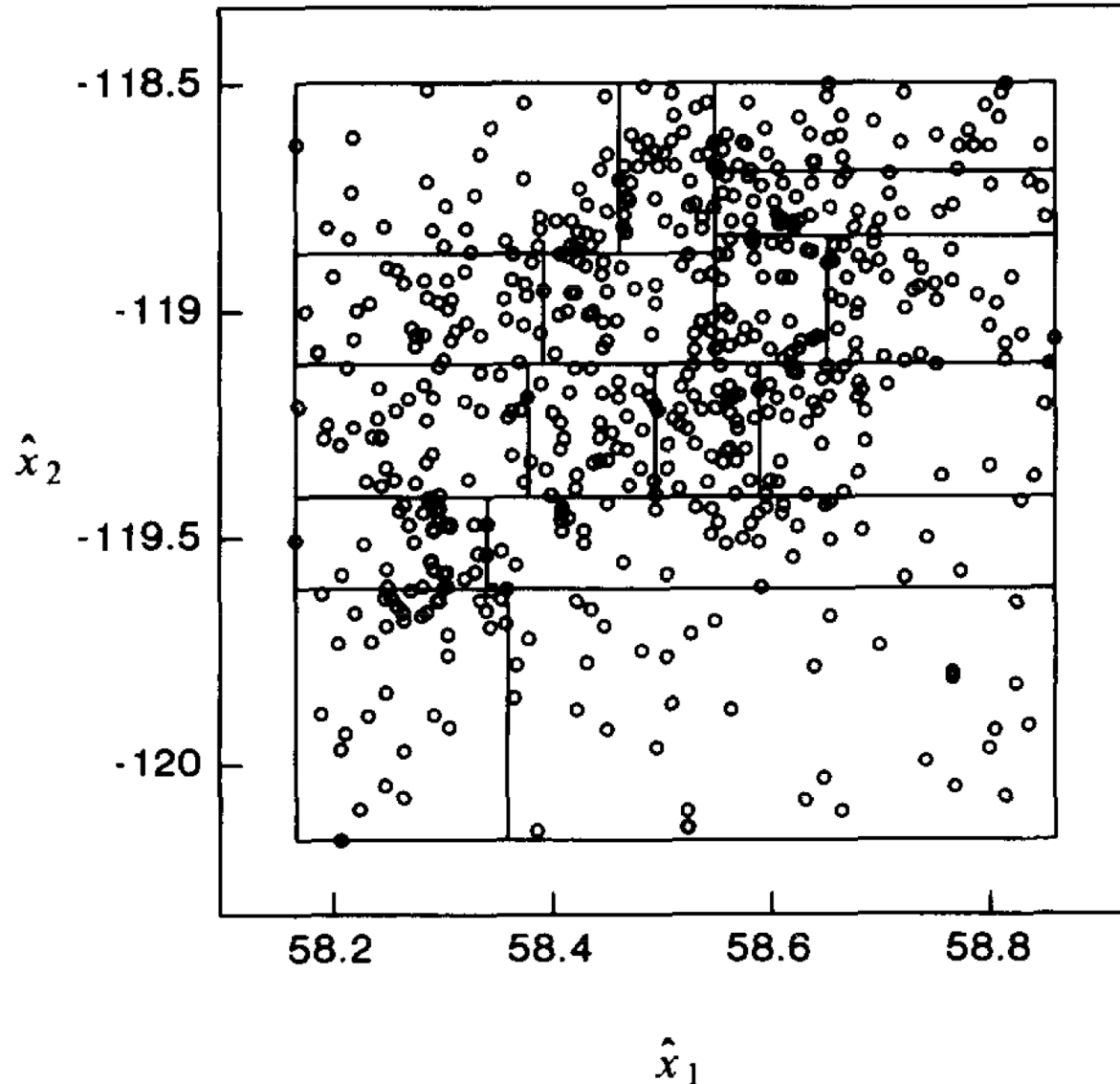
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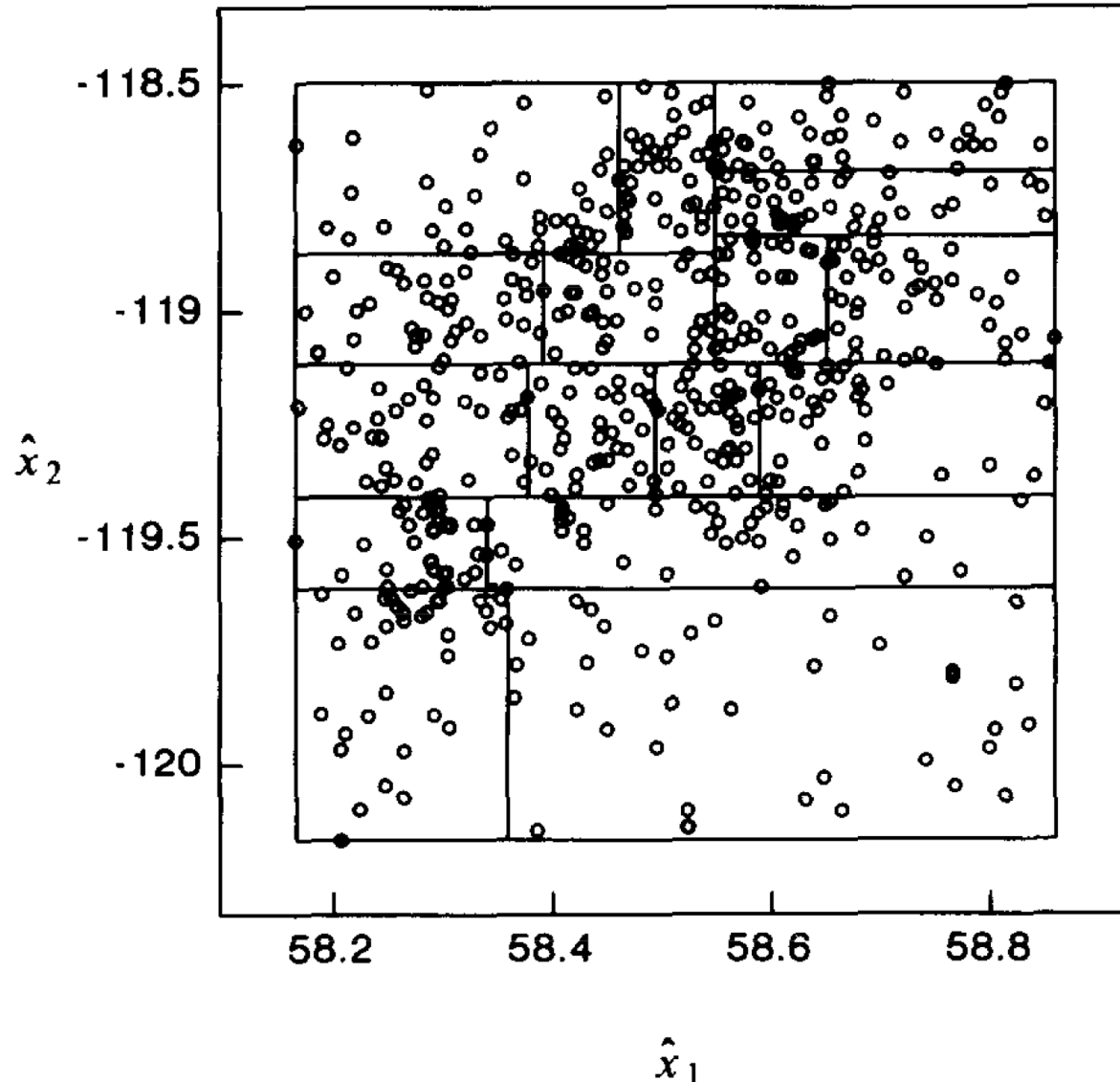


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Fit a spatial tree on **feature** data
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“Cluster” on X alone

WHAT IS CLUSTERING REGRESSION



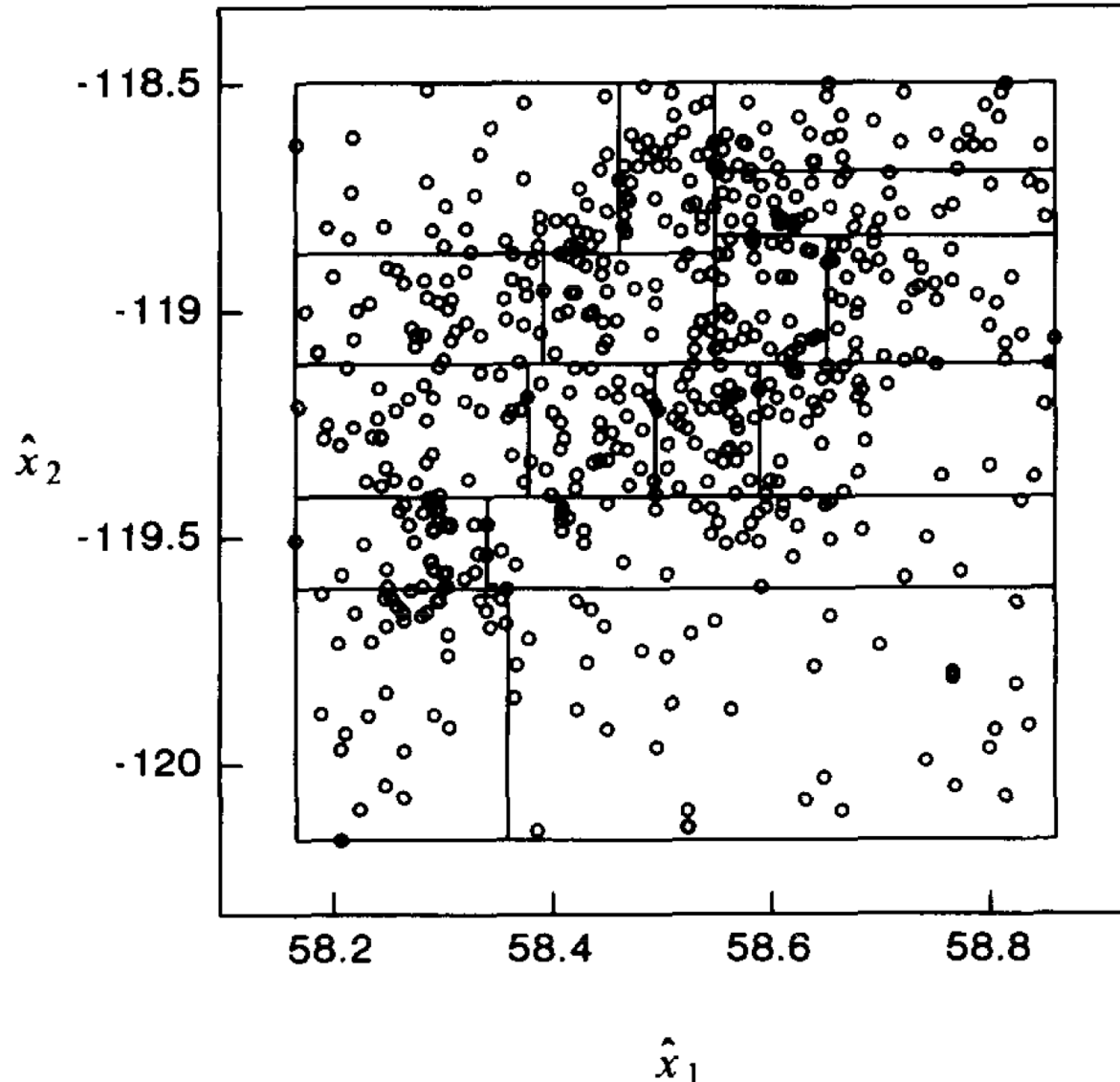
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**Tree uses location,
not regression loss**

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LOWESS Idea:

Fit a **spatial tree** on **feature** data
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“Cluster” on X alone

**Tree uses location,
not regression loss**

**Can we
do better?**

WHAT IS CLUSTERING REGRESSION

CLUSTERING REGRESSION

jointly solving clustering and regression

(GEO)CLUSTERING REGRESSION

long-studied, quite recently revived

UNDERSTANDING QUADTREES

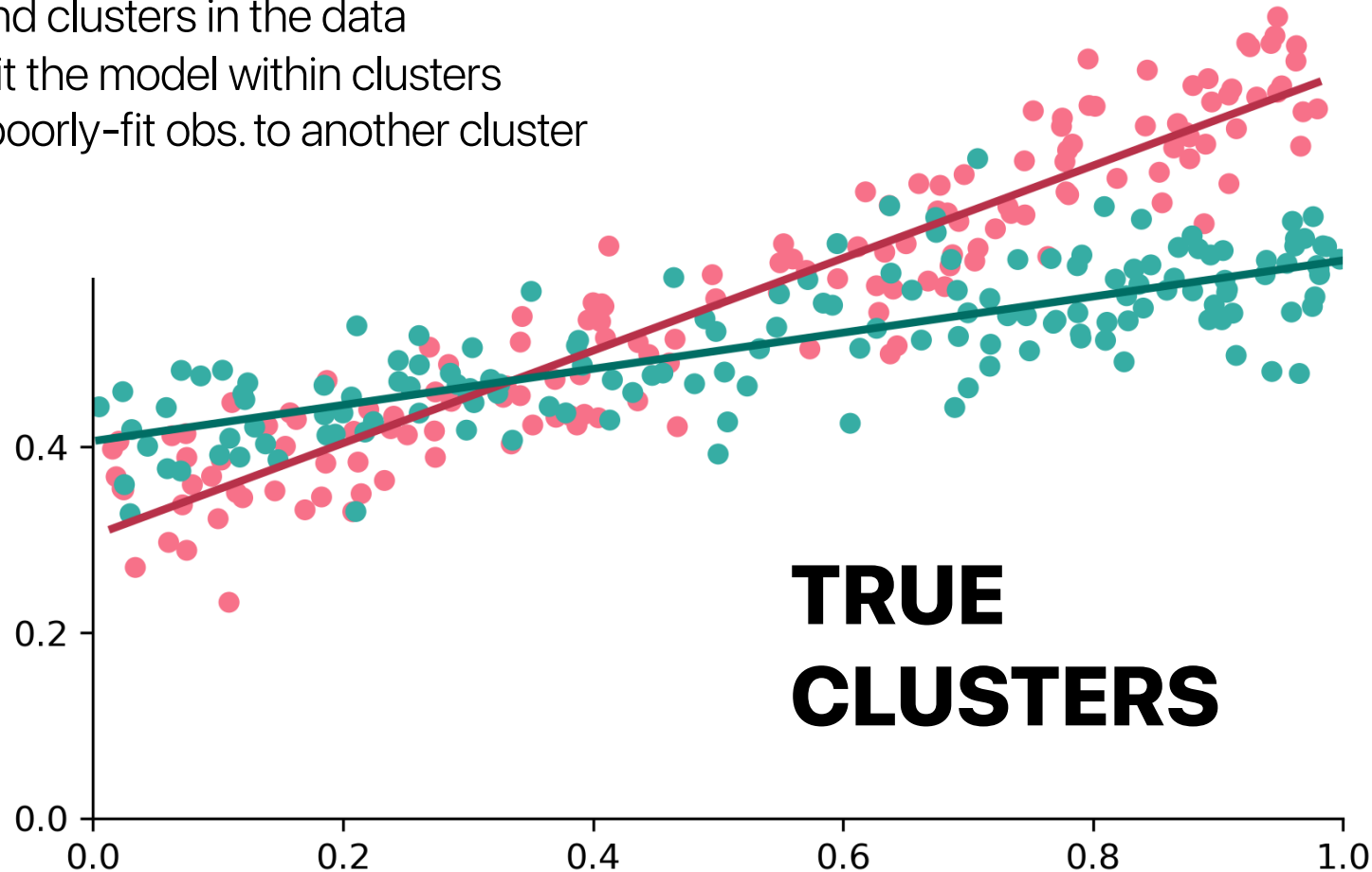
APPLYING QUADTREE REGRESSION

THE SUPERVISED QUADTREE

Optimized ESS: 198.1572, True ESS: 224.6662

Späth (1979)'s three-stage process

- **CLUSTER:** find clusters in the data
- **ESTIMATE:** fit the model within clusters
- **REFINE:** flip poorly-fit obs. to another cluster

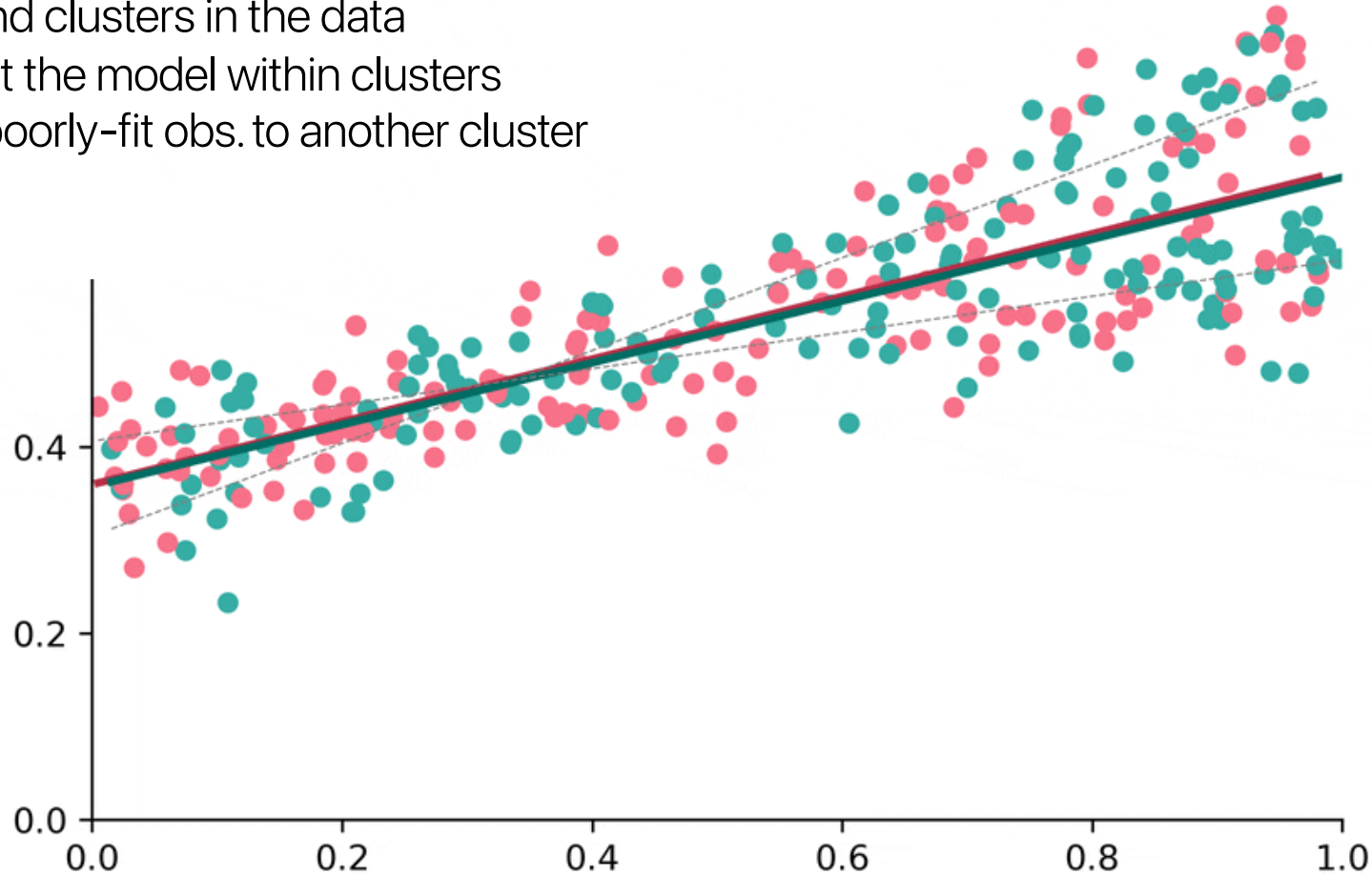


SPÄTH (1979) ITERATIVE CLUSTERING

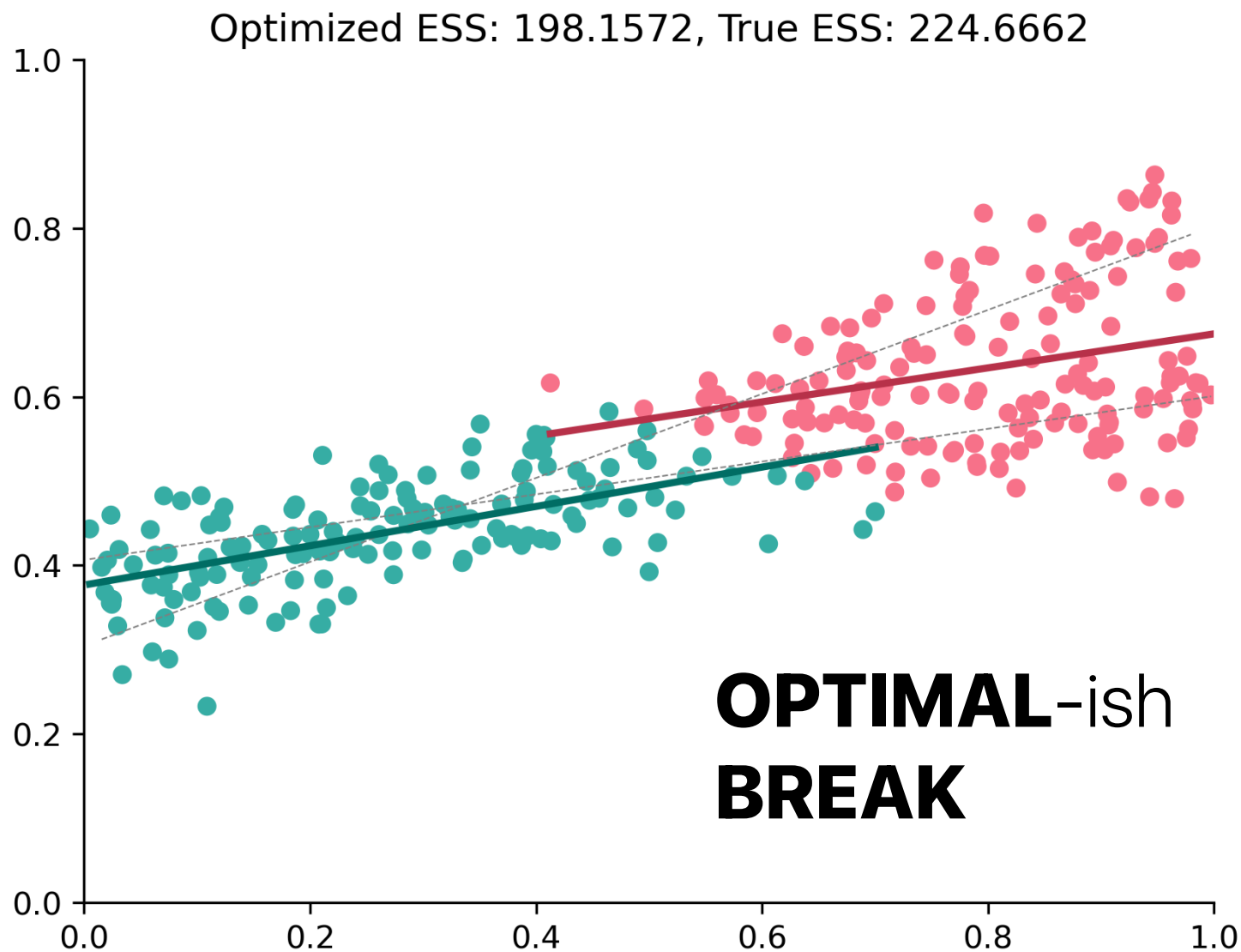
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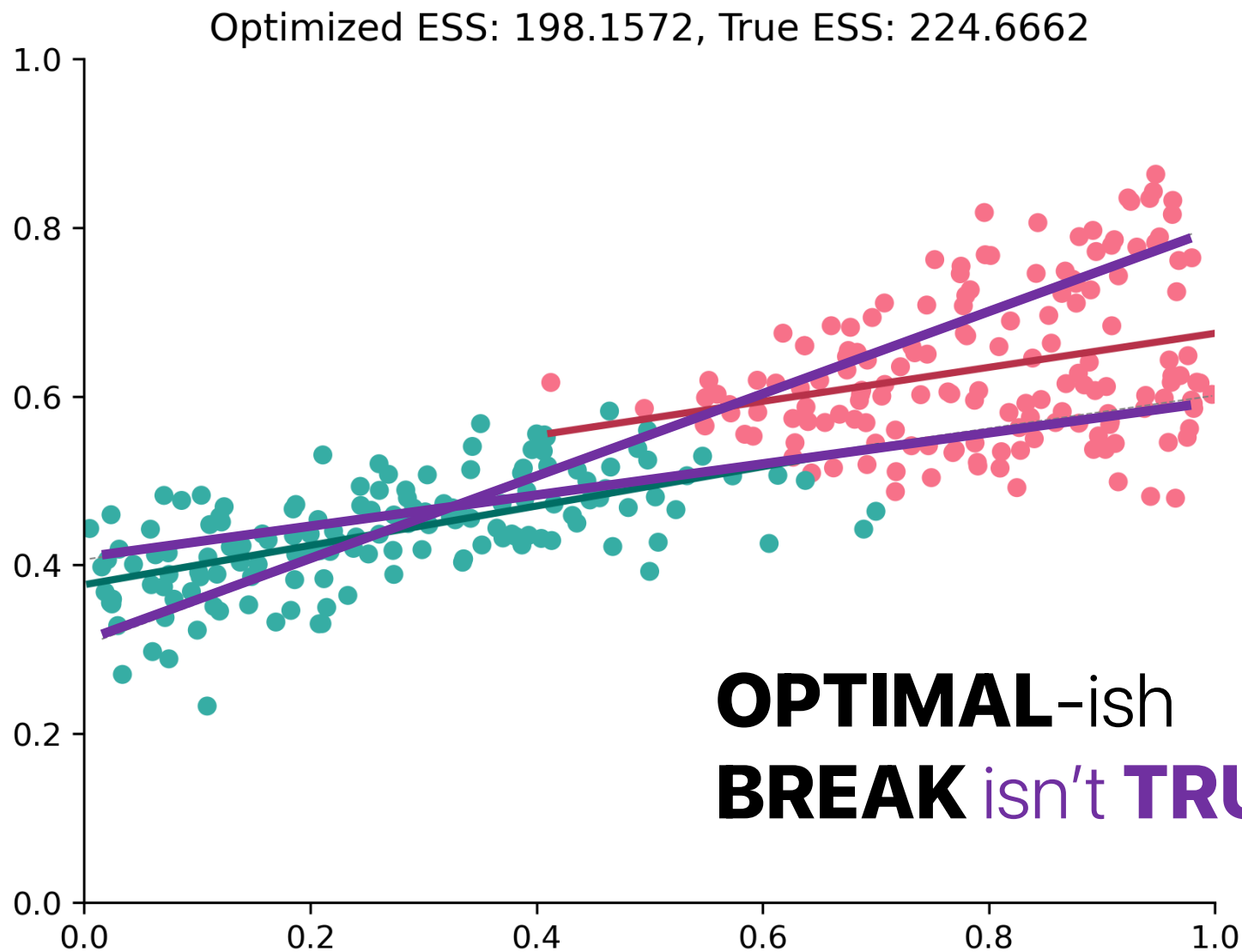
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OPTIMAL-ish
BREAK isn't **TRUE**

SPÄTH (1979) ITERATIVE CLUSTERING

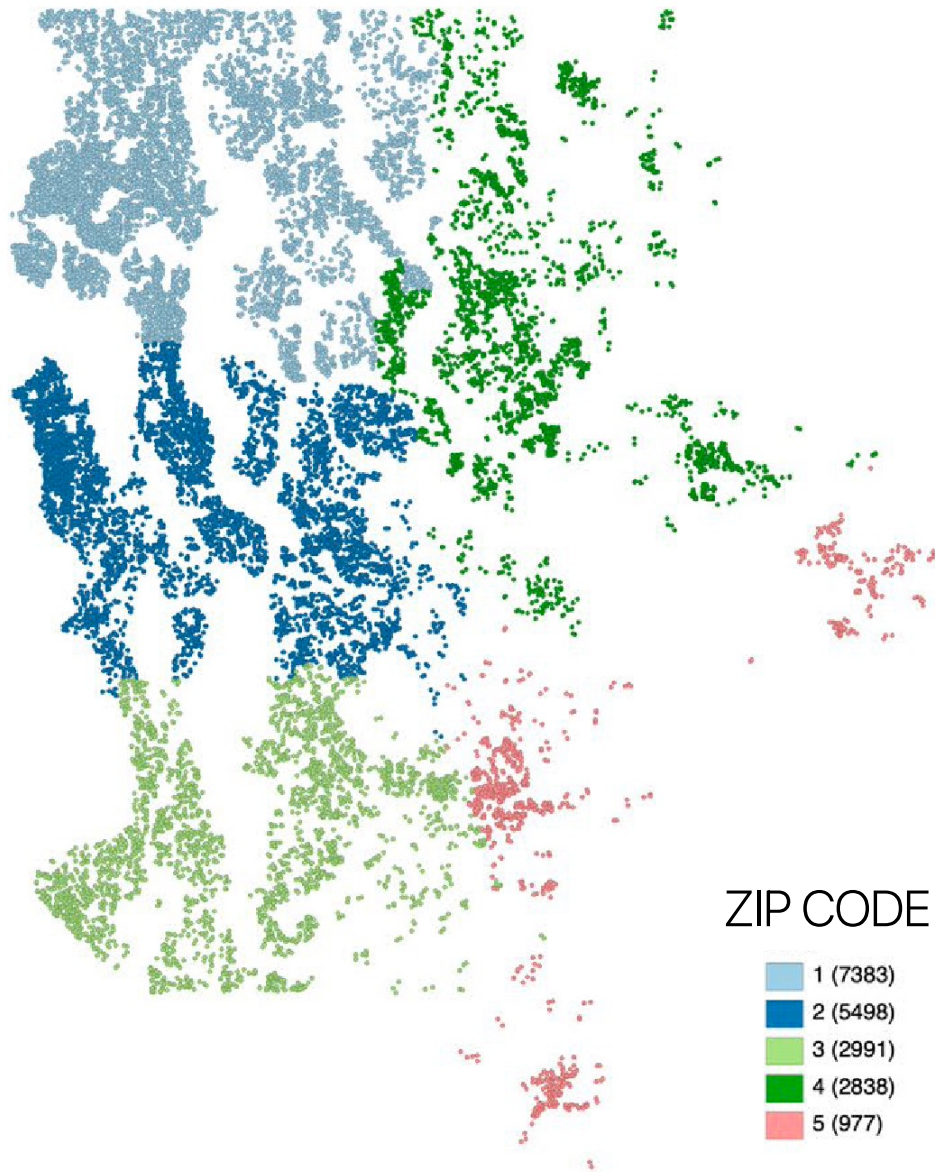
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- **CLUSTER:** find clusters in the data
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Heuristic (i.e. not optimal), slow+computationally intensive, even in the seriously improved spatial variant provided by Sugawara and Murakami (2021)
No uncertainty for checking assignment, hence no guarantee of recovering the true classes

BEFORE THE SUPER*TREE LEARNER

Spatial SUR Models



WHAT IS CLUSTERING REGRESSION

Spatial SUR Models

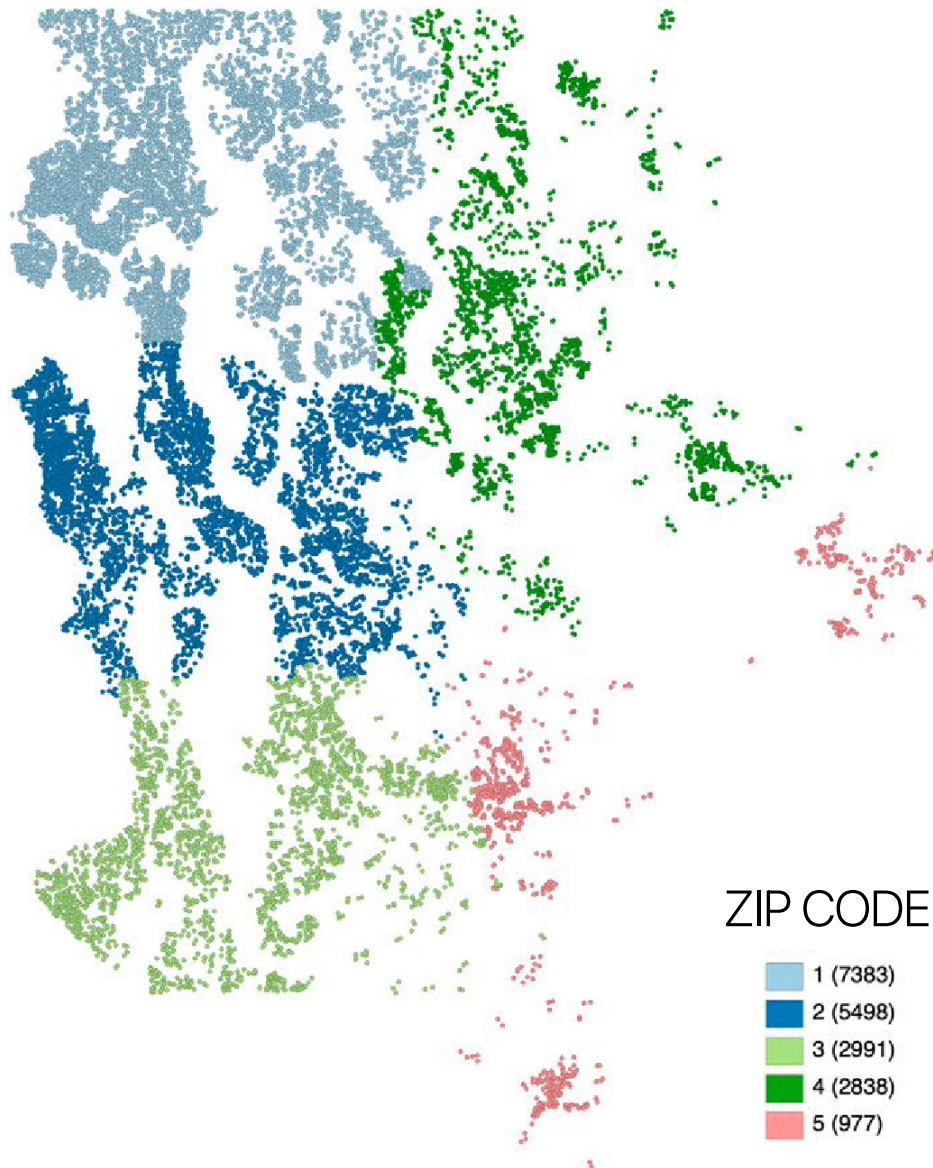
Stack each group's $y \sim X$ model together

JOURNAL OF REGIONAL SCIENCE, VOL. 30, NO. 2, 1990, pp. 185–207

**SPATIAL DEPENDENCE AND SPATIAL STRUCTURAL
INSTABILITY IN APPLIED REGRESSION ANALYSIS***

Luc Anselin†

Test to check if separate models are needed!



WHAT IS CLUSTERING REGRESSION

Spatial SUR Models

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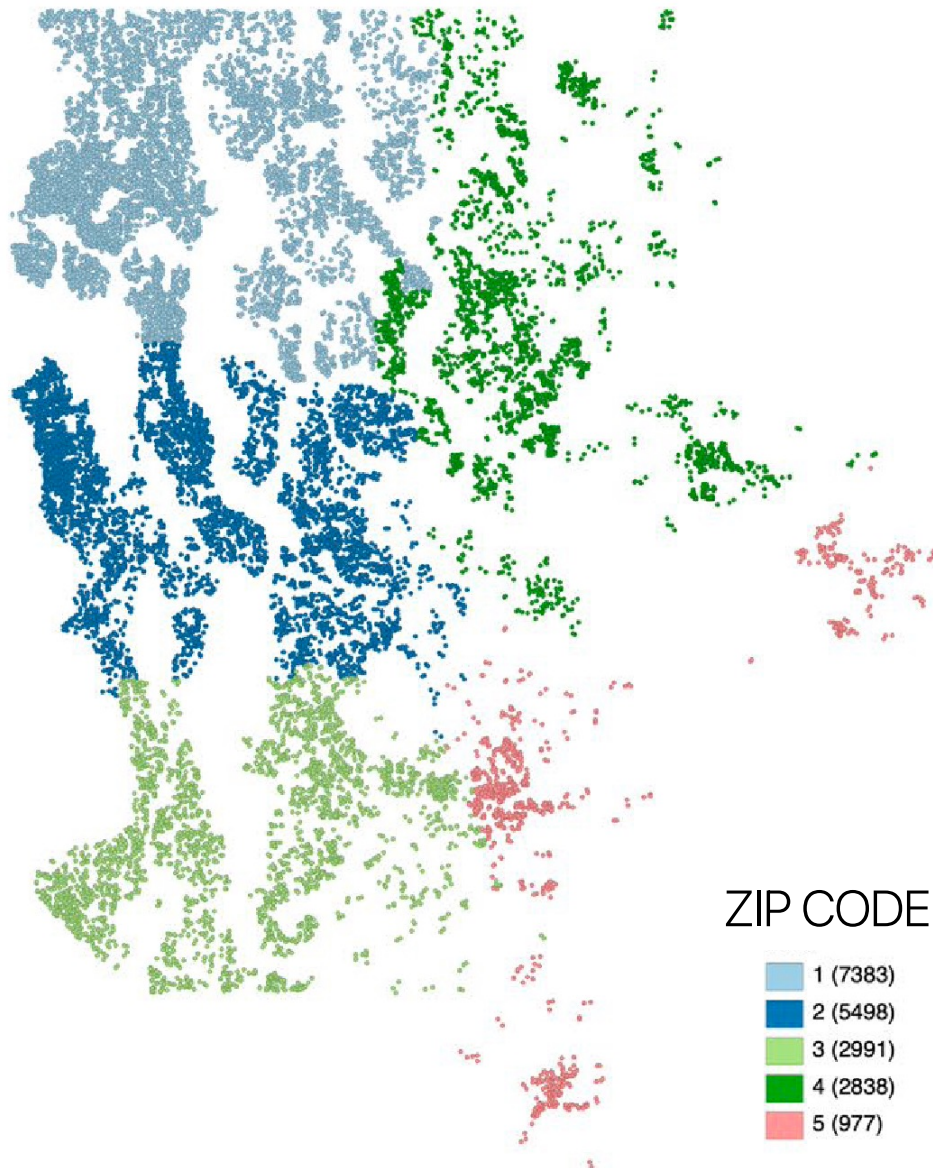
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Test to check if separate models are needed!

Can we integrate *that* test during fitting?



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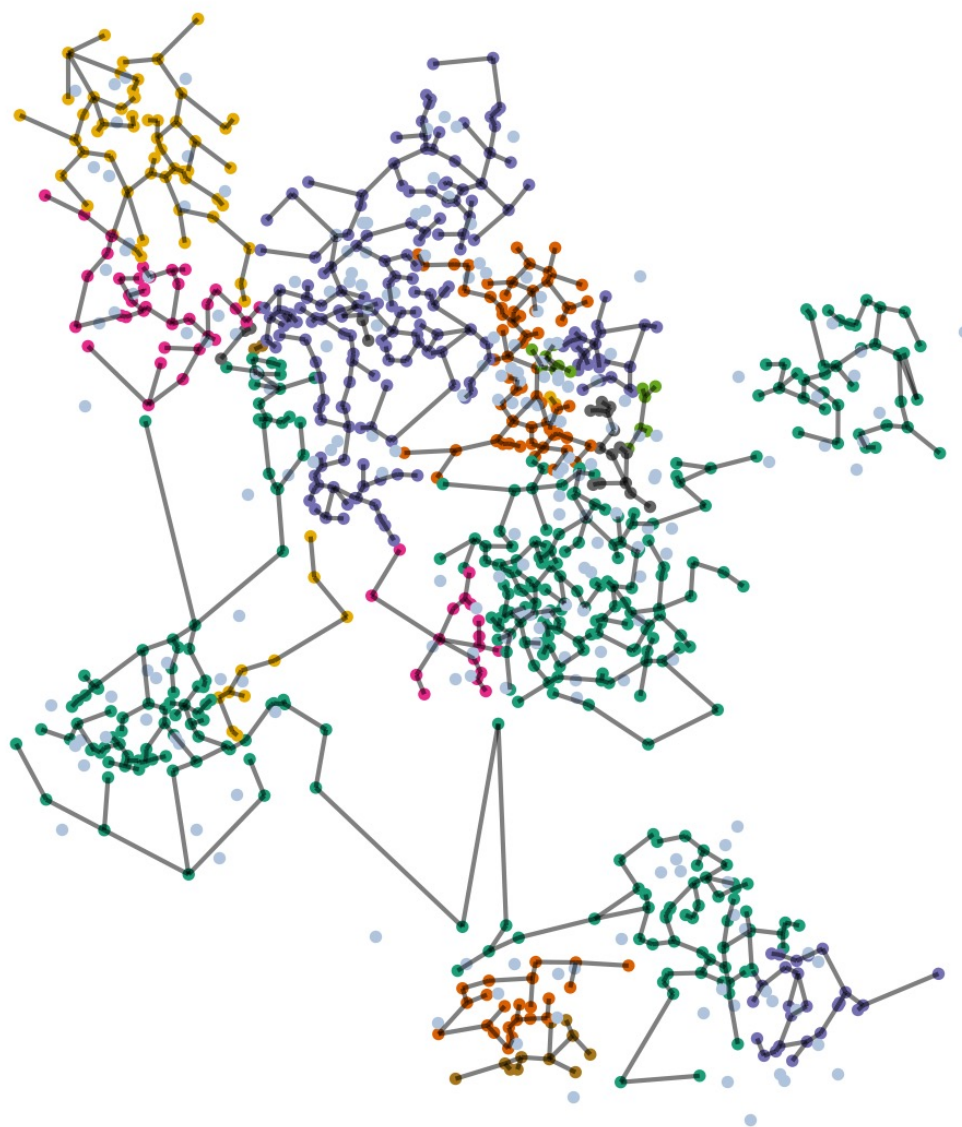
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No guarantee of recovering the original classes

Anselin & Amaral (2023)

- **COMBINE:** Combine spatial and feature similarity
- **SPAN:** compute a MST over the combined affinity matrix
- **PRUNE:** remove MST links to create clusters according to Chow test on $(SSR - (SSR_a + SSR_b))$

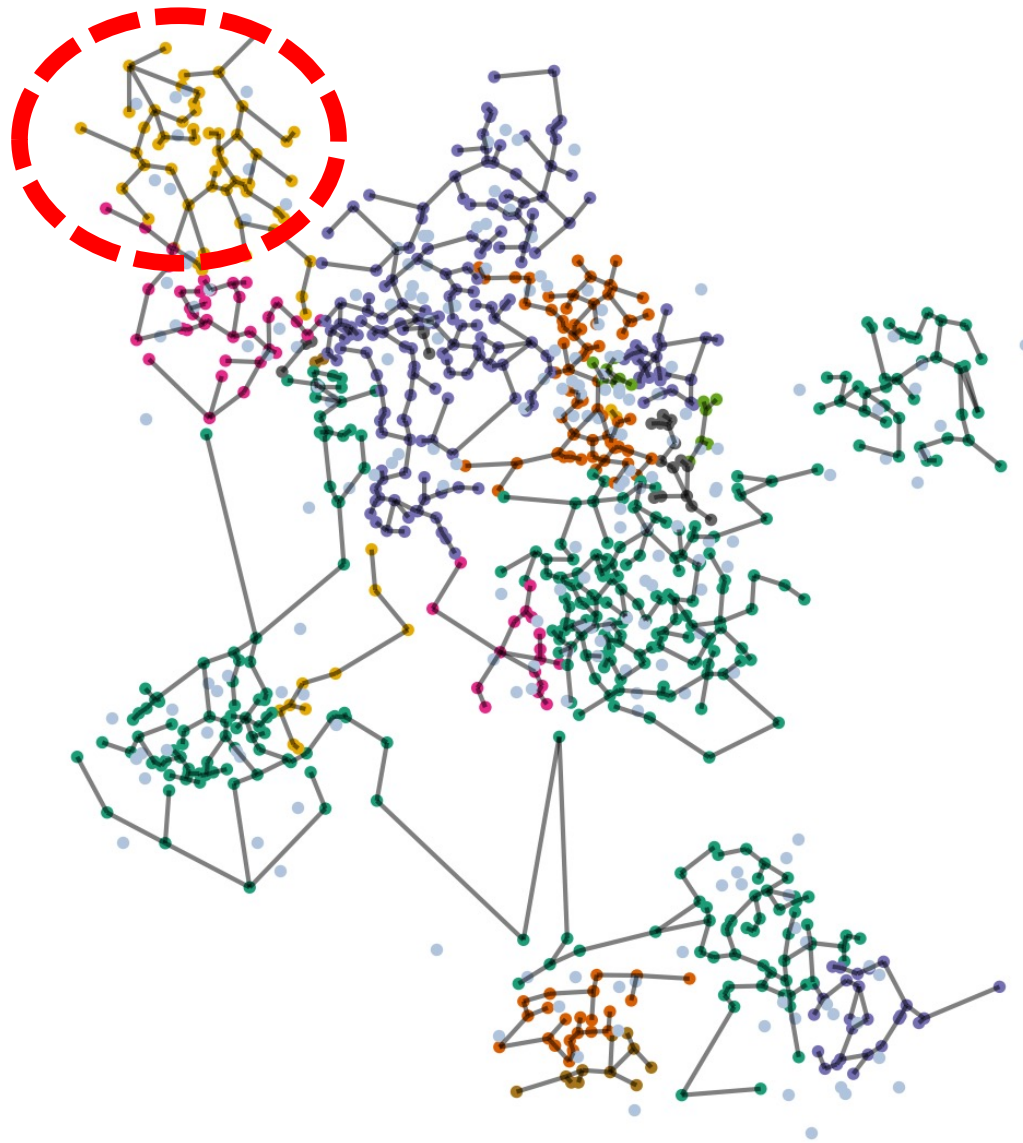
BEFORE THE SUPER*TREE



SKATER REGRESSION

each cluster gets
one regression

Clusters are subtrees
with good regressions.



SKATER REGRESSION

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SKATER Additive Learner (STAKERAL): A Data-driven Method for Delineating Covariate-specific Endogenous Spatial Regimes

Sui Zhang^{*1}, Wei Zheng^{†1} and Cecilia Wong^{‡1}

¹Spatial Policy and Analysis Lab, Department of Planning, Property and Environmental Management, the University of Manchester

GISRUK 2025

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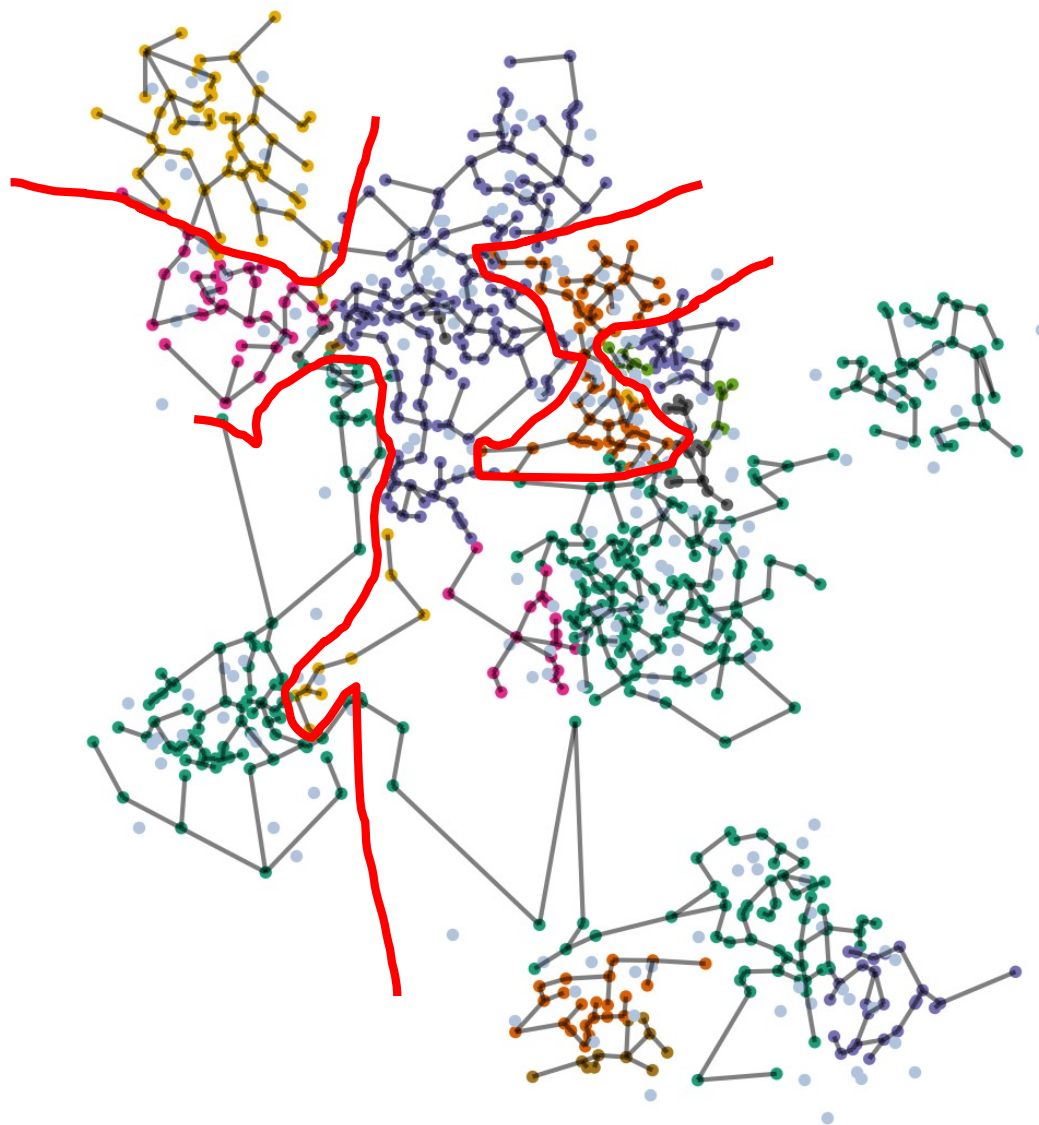
BEFORE THE SUPERVISED QUADTREE

each cluster gets
one regression
dummy term

stack these into a
dummy matrix, **D**

Regression becomes
 $y \sim \mathbf{D} + \mathbf{DX}$

clusters are regions
where interaction term
improves model fit

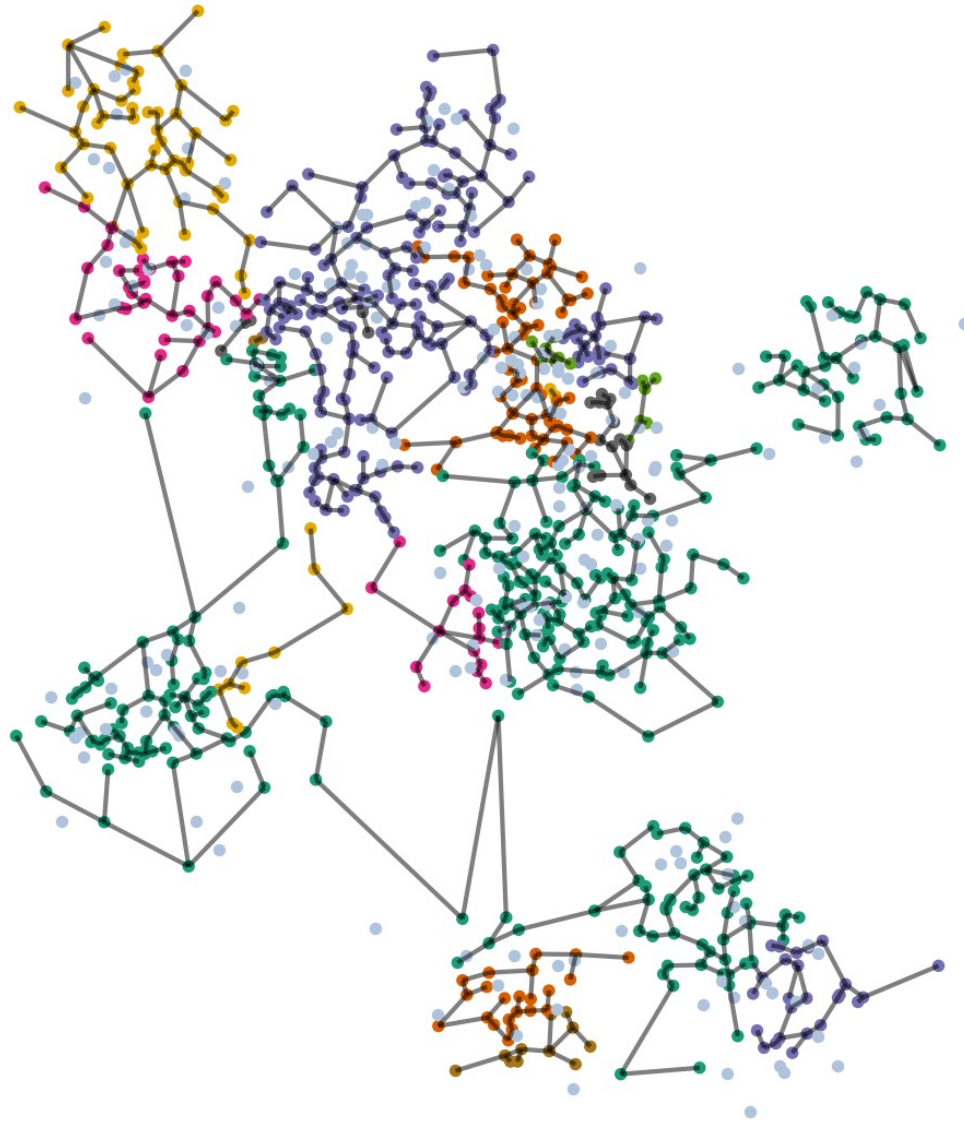


SKATER REGRESSION

Strengths:

Judicious use of degrees of freedom

Uses LRT to find good clusters



SKATER REGRESSION IN PRINCIPLE

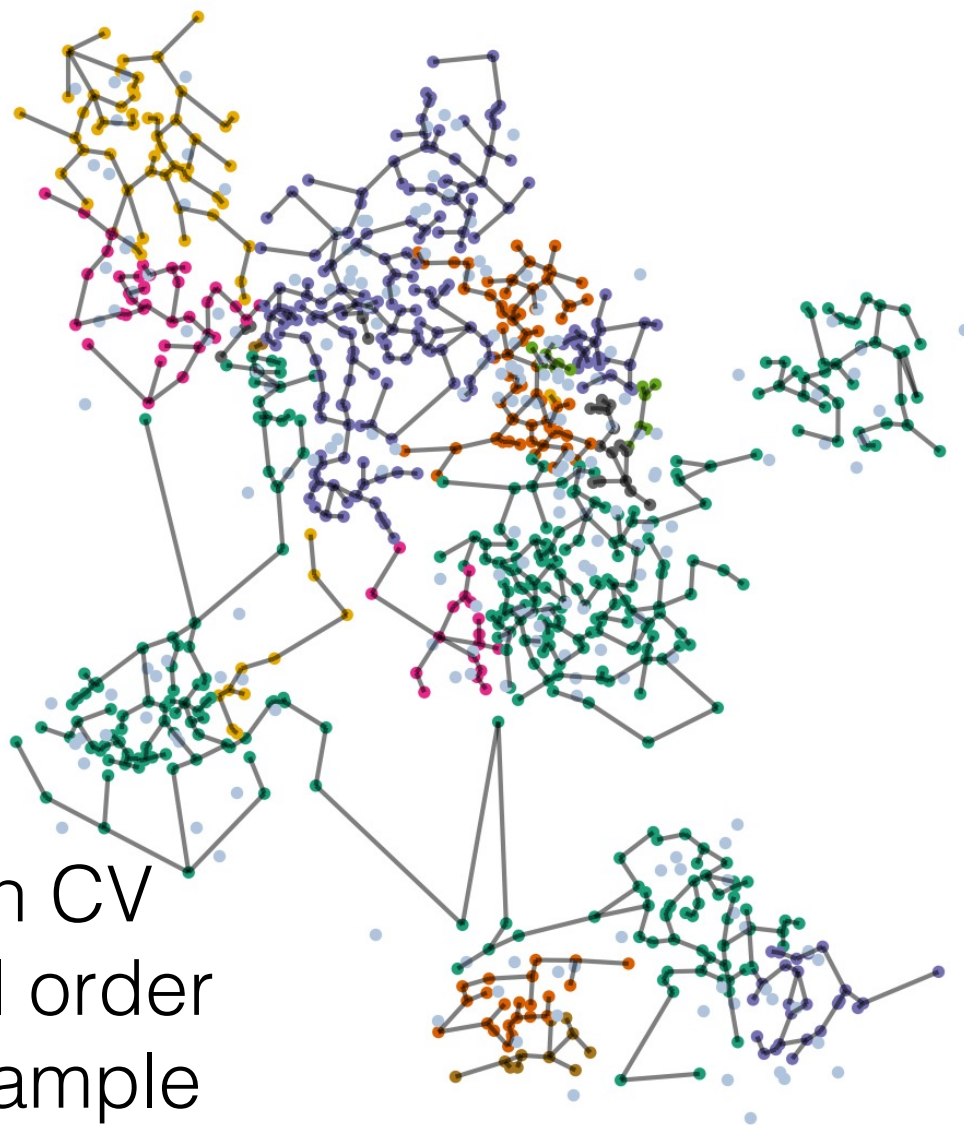
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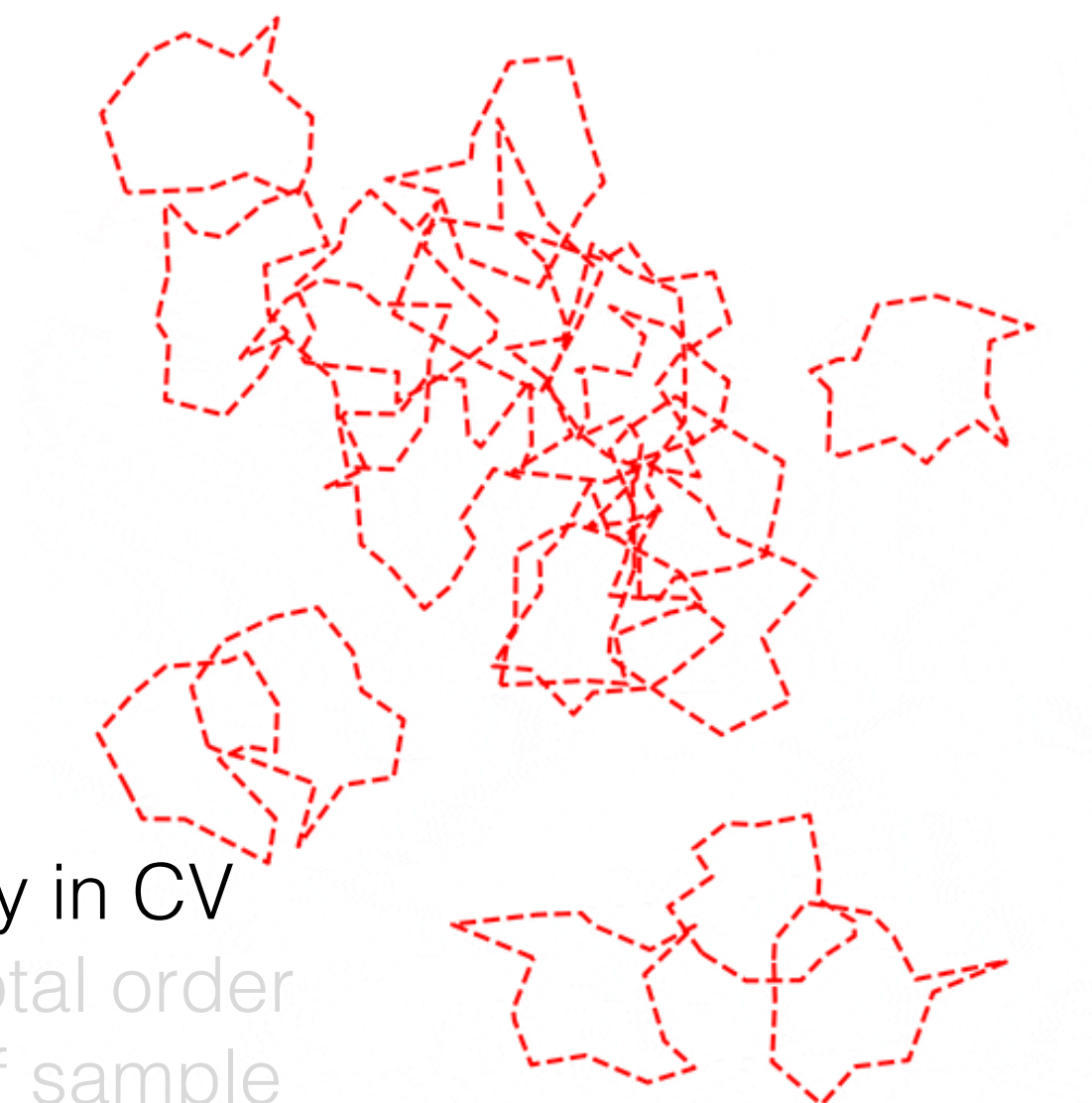
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Weaknesses:

- MST is finnickyy in CV
- MST has no total order
- can't do out of sample



SKATER REGRESSION IN PRINCIPLE



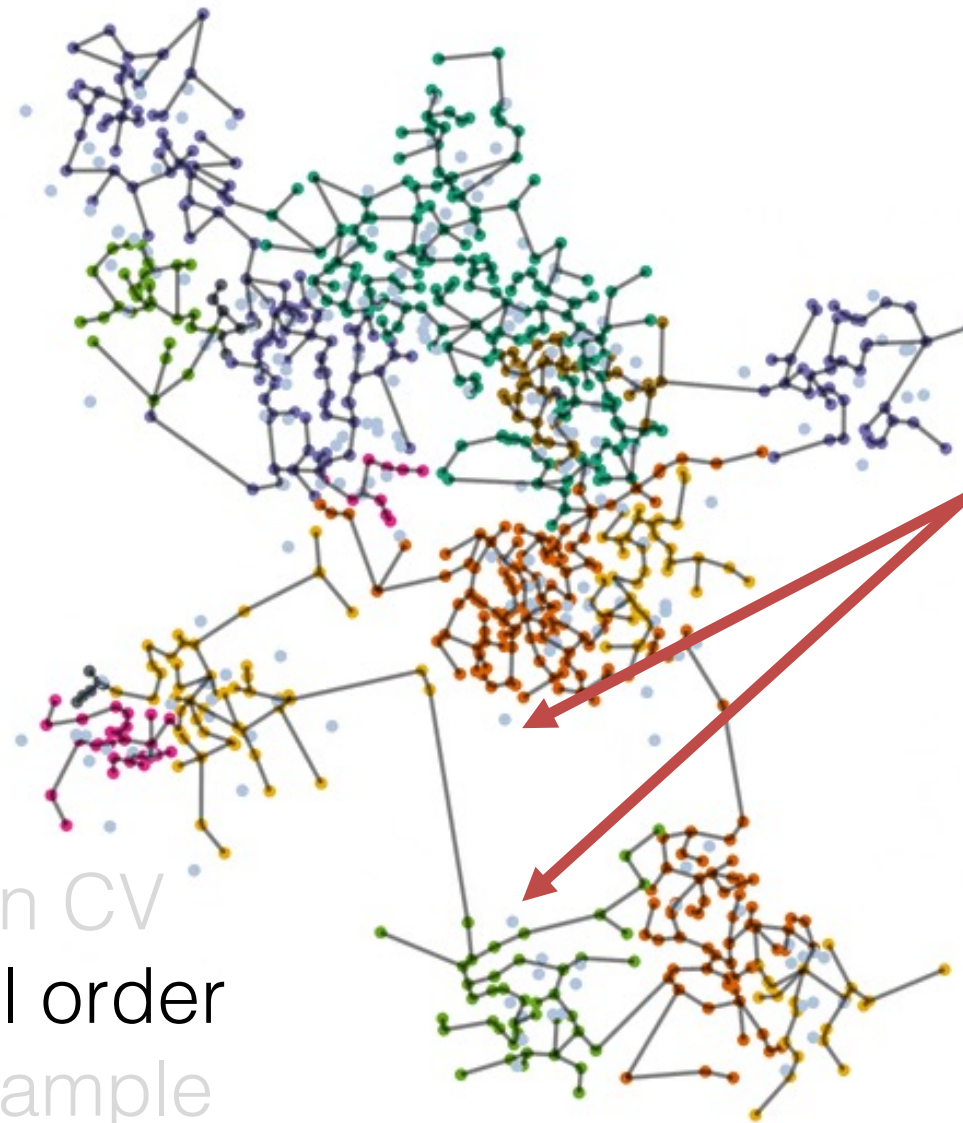
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MSTs ARE VERY SENSITIVE OBJECTS

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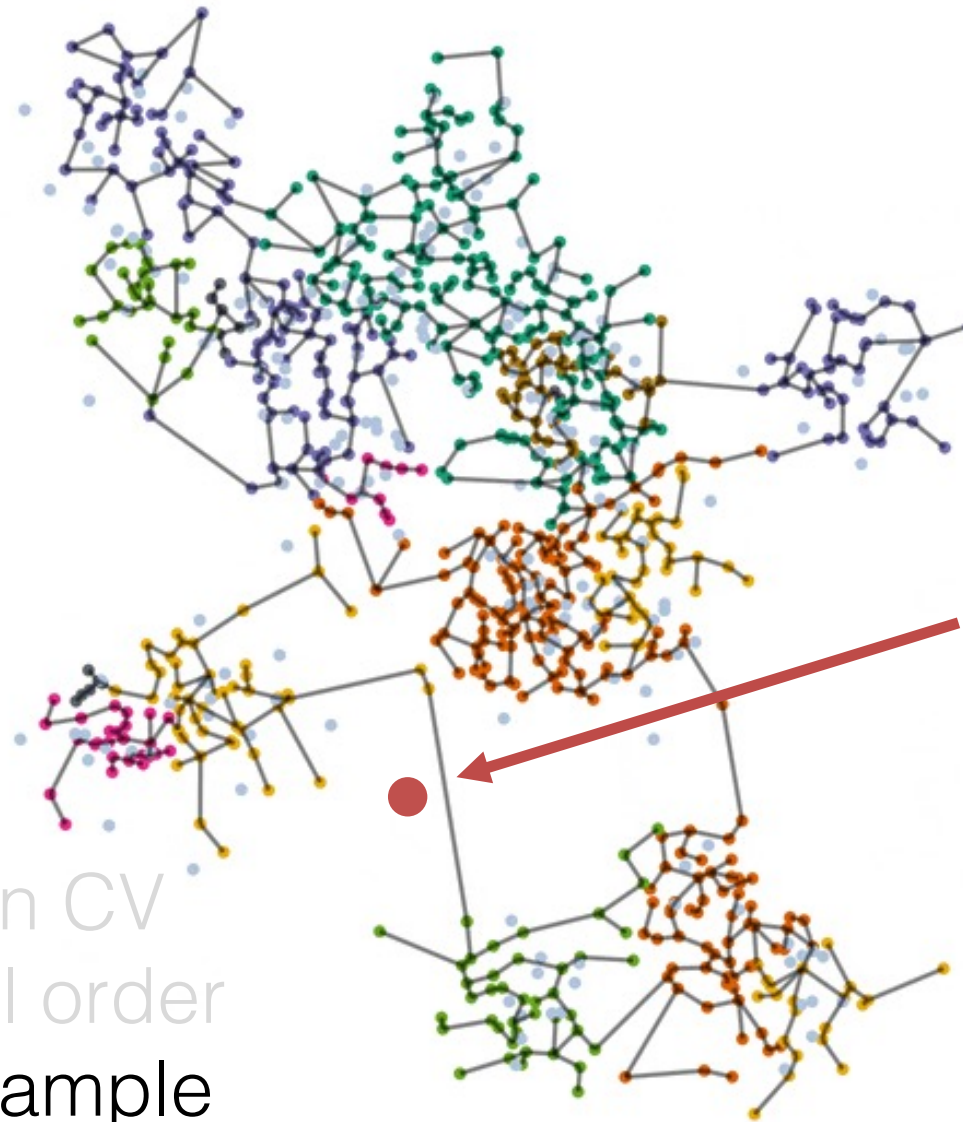


Very distant in
some splits,
neighbors in others!

TREE ADJACENCY IS NOT SIMILARITY

Weaknesses:

- MST is finnick in CV
- MST has no total order
- can't do out of sample



How should this connect?

Nearest leaf?
Nearest node?
Re-tree everybody?

NEW POINTS BREAK MST STRUCTURE

CLUSTERING REGRESSION

jointly solving clustering and regression

GEOGRAPHICAL CLUSTER-REG

jointly solving clustering and regression

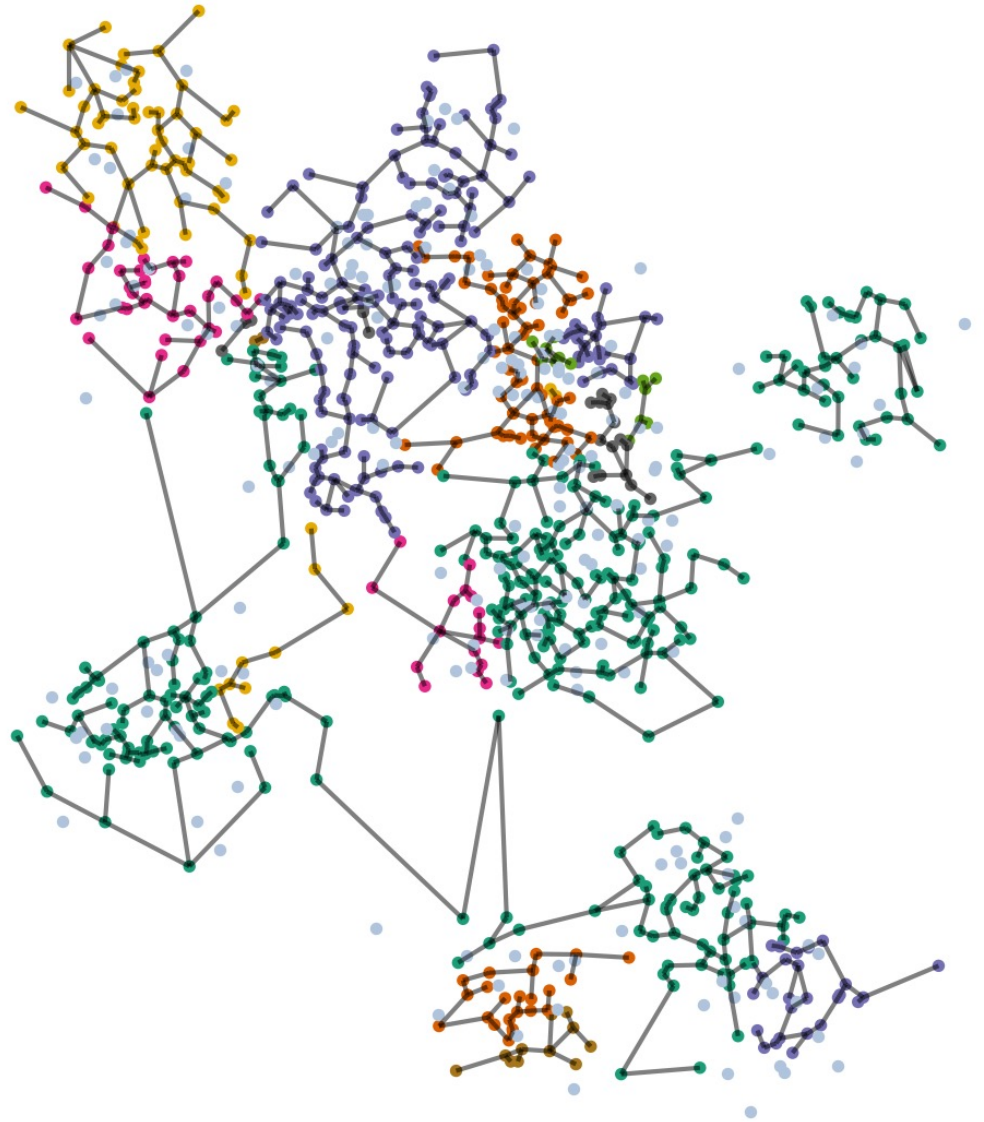
UNDERSTANDING SUPER*TREES

spatial splits for spatial fits

APPLYING QUADTREE REGRESSION

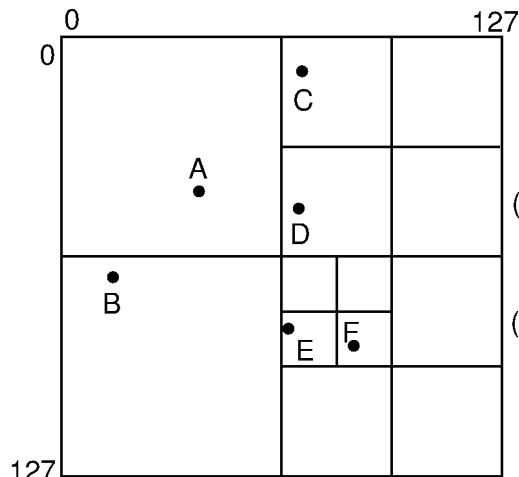
THE SUPER*TREE REGRESSION

Build a Minimum Spanning Tree
Split tree when it improves predictions
Aggregate subtrees to make predictions

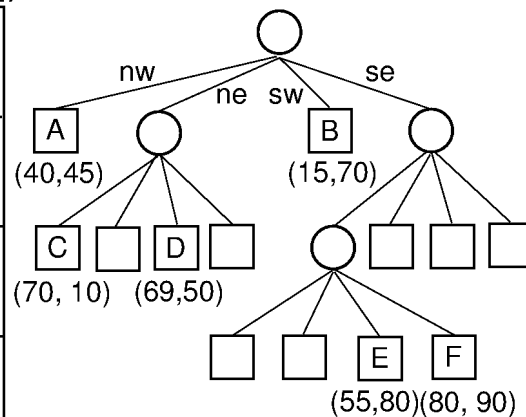


BEFORE THE SUPER*TREE

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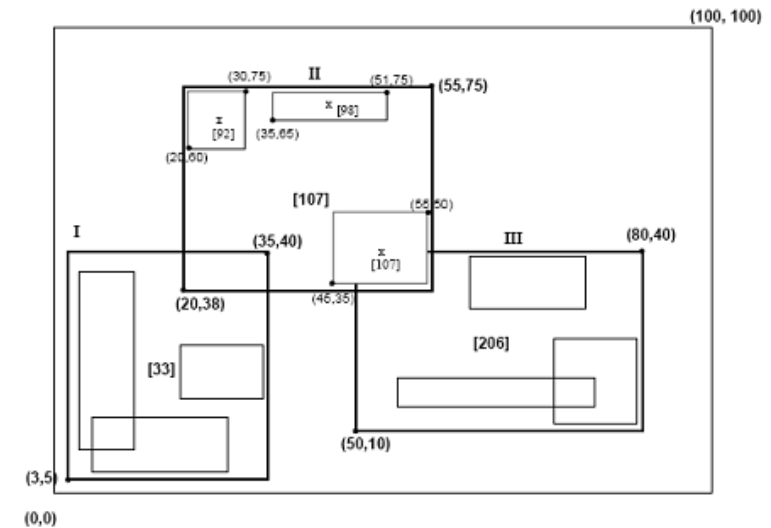


(a)

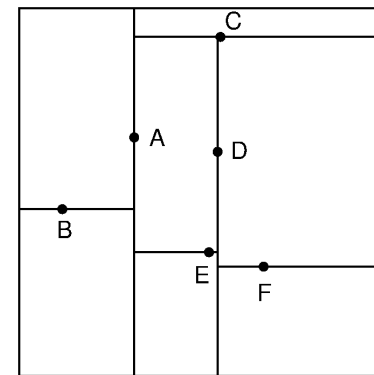


(b)

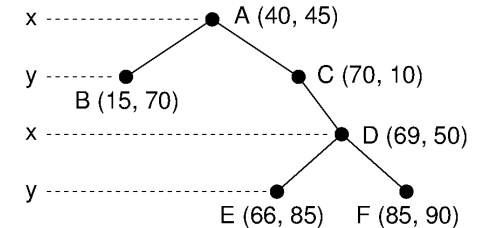
QuadTree



Hilbert R-Tree



(a)

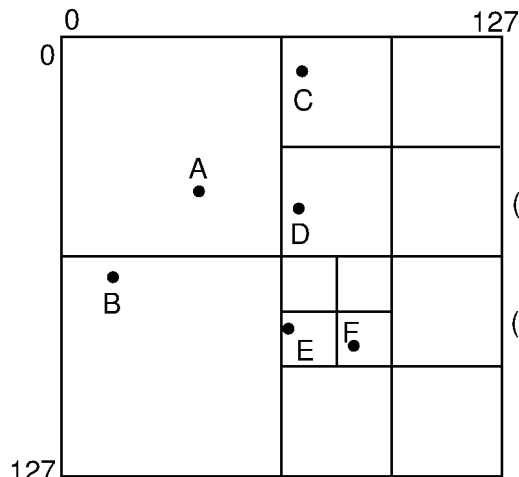


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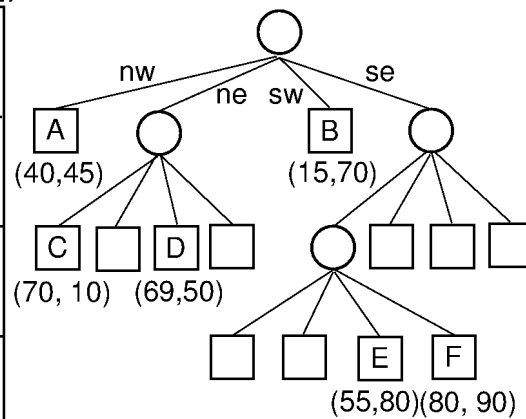
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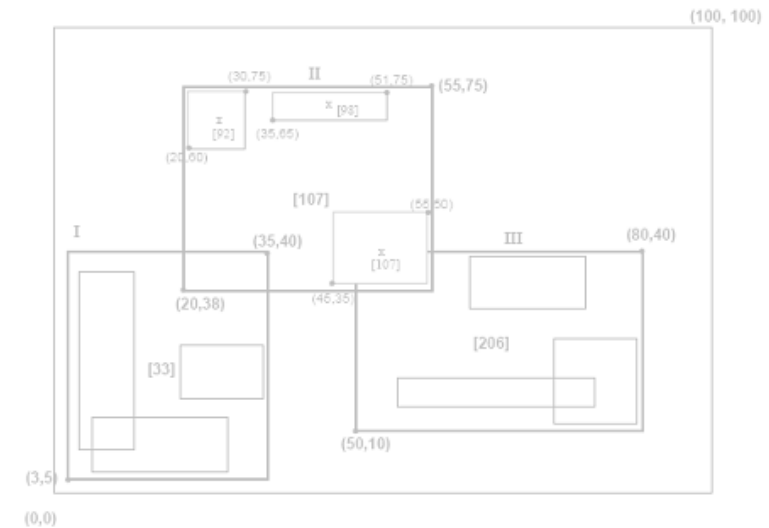


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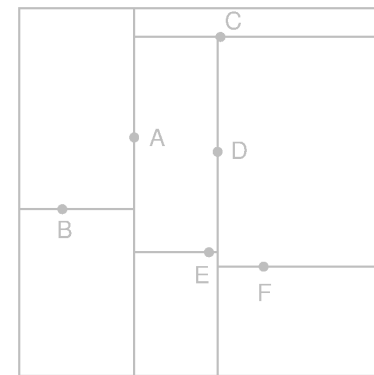


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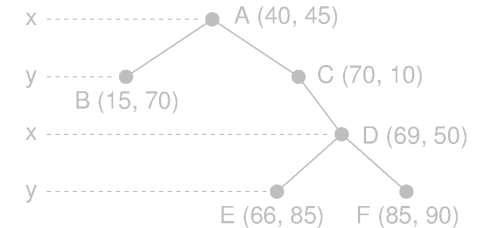
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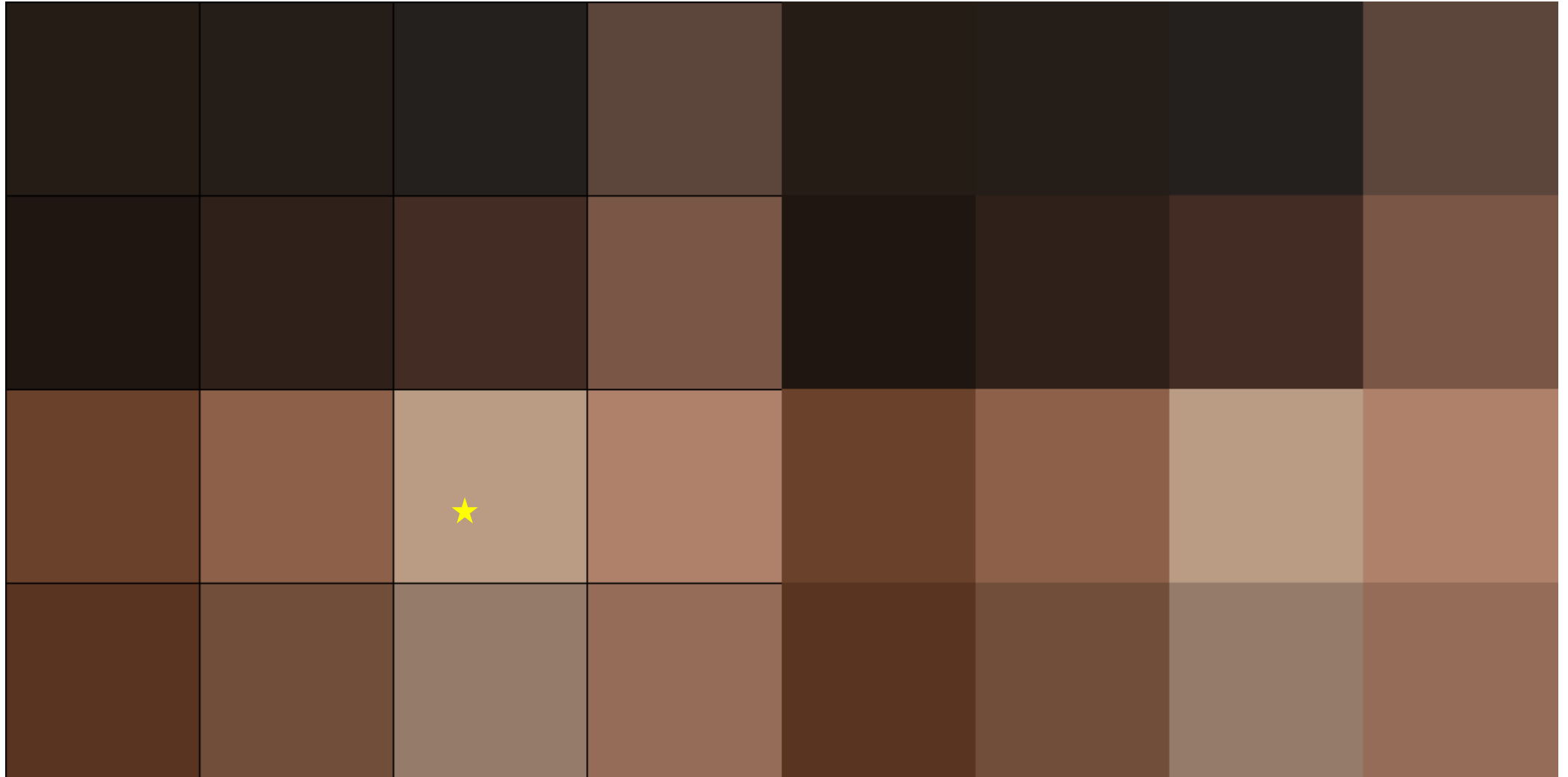
(b)

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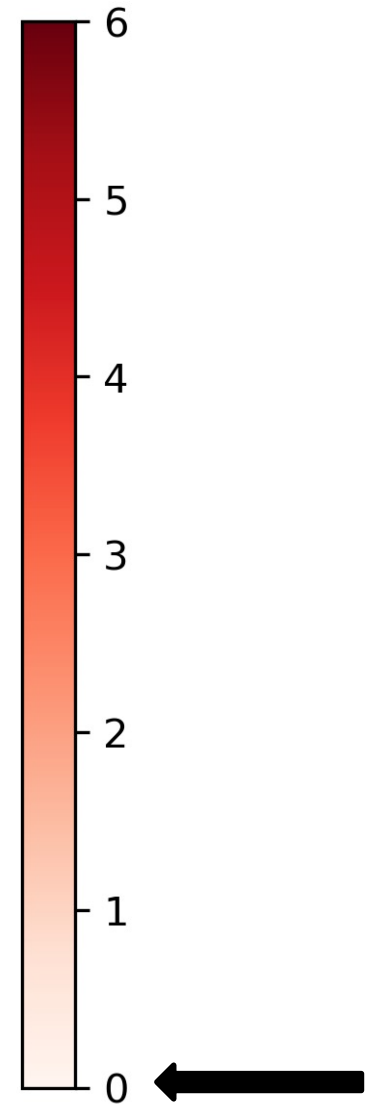
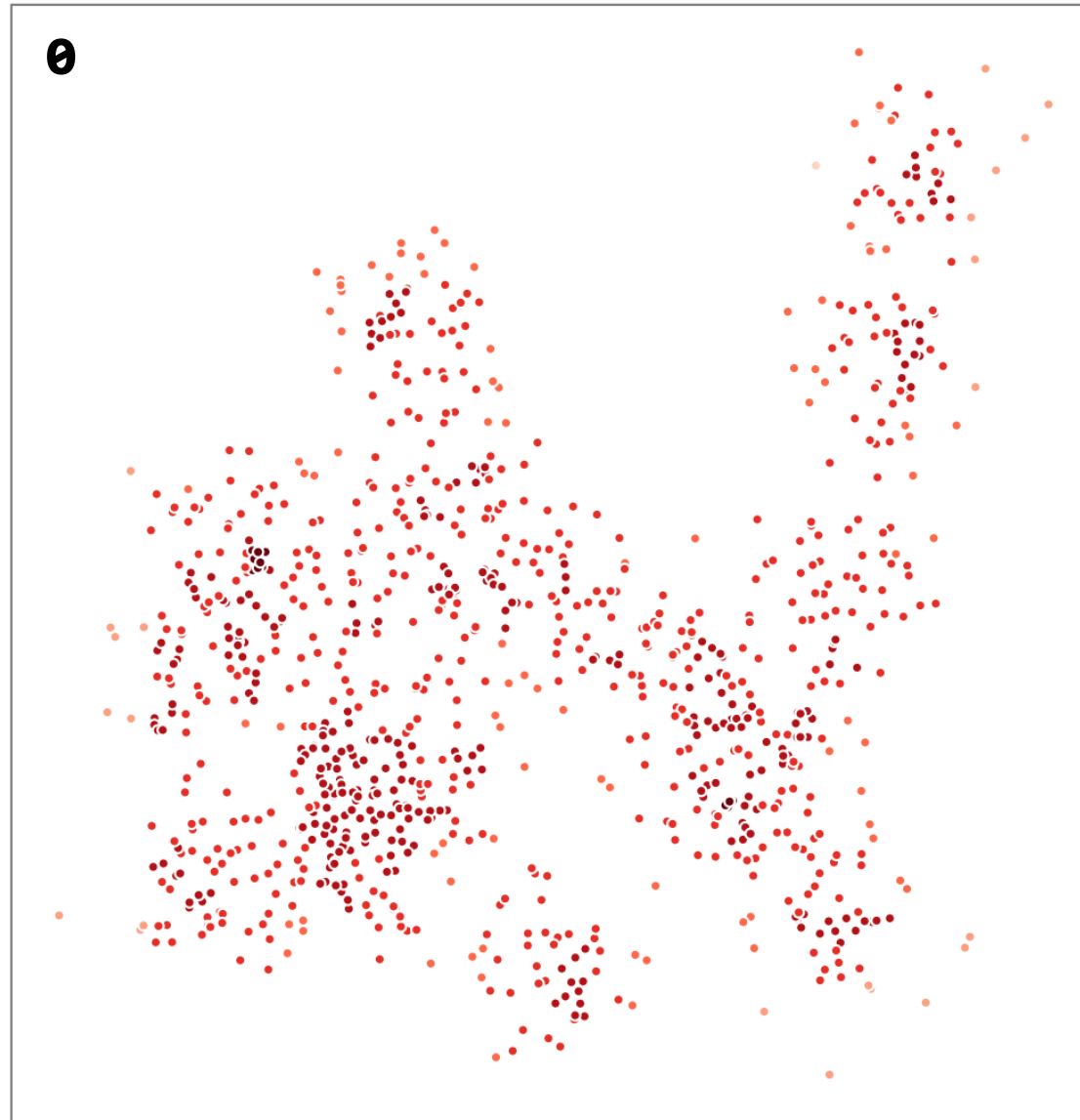
THE SUPER QUADTREE (SUPERQT)



QUADTREES IN IMAGE COMPRESSION

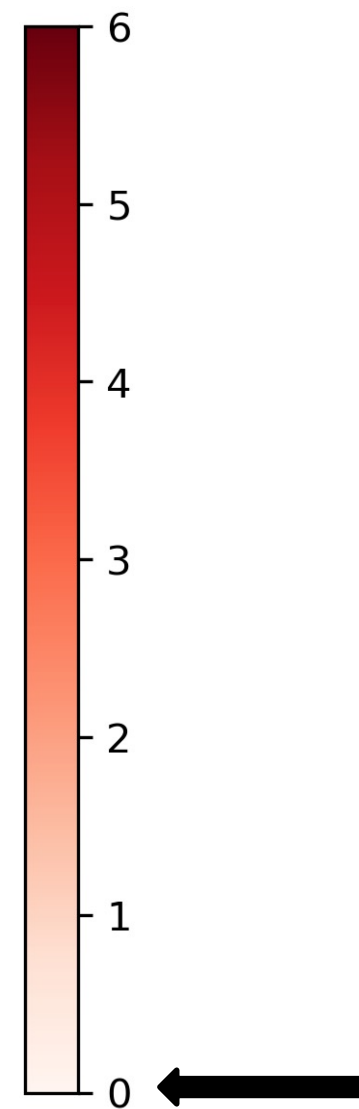
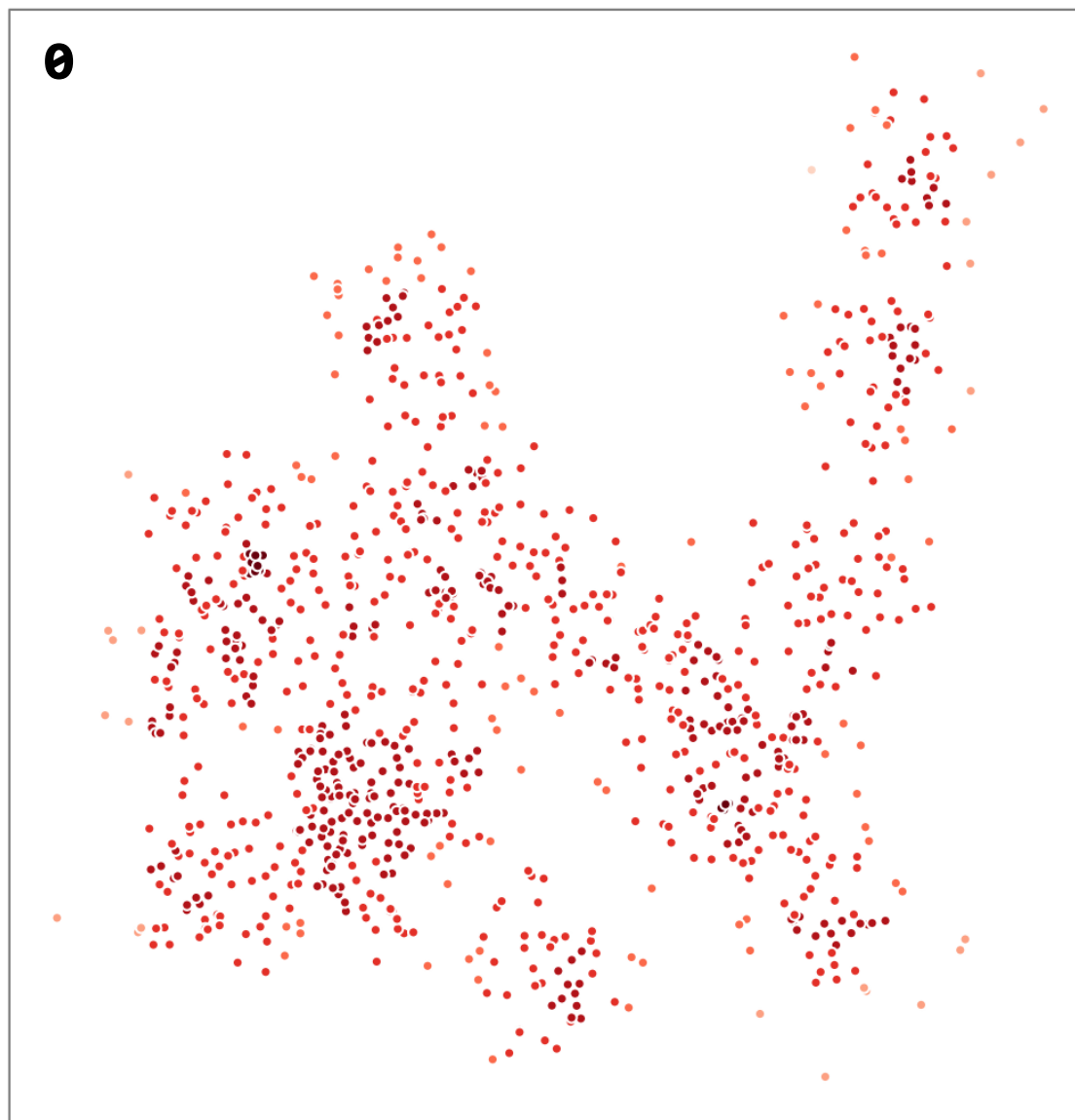


SPEND DETAIL WHERE ITS NEEDED MOST

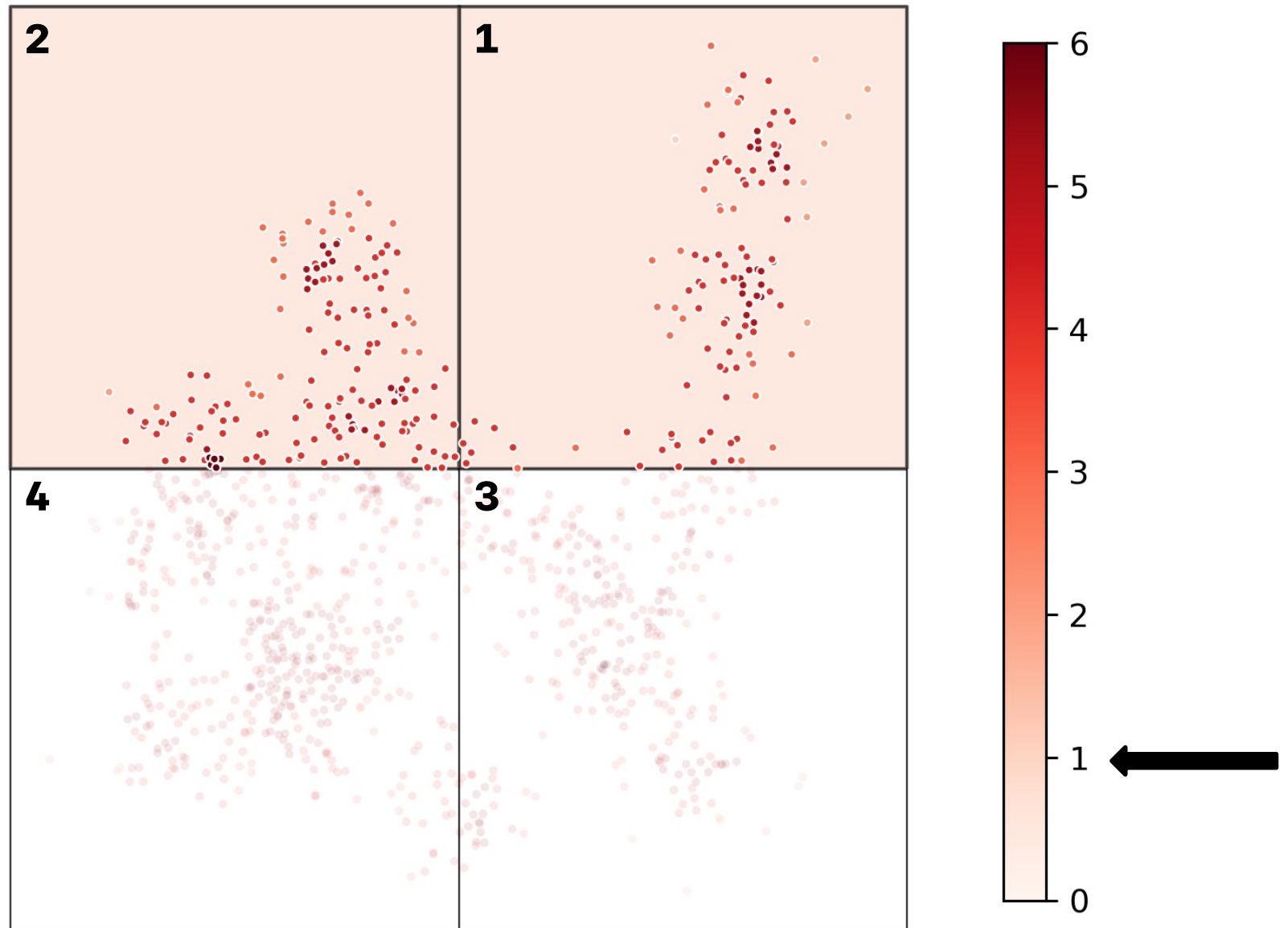


WALK DOWN THE DEPTH LADDER

c	x_1
1	x_{11}
1	x_{12}
1	x_{13}
1	x_{14}
1	x_{15}
...	...
1	x_{1n}

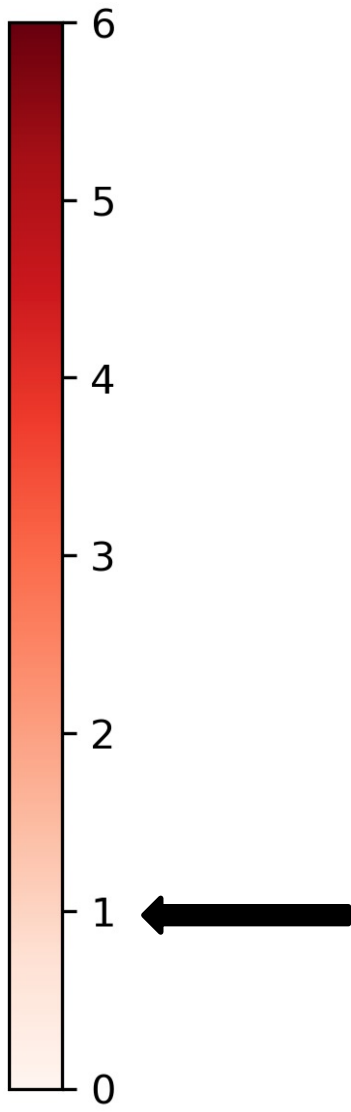
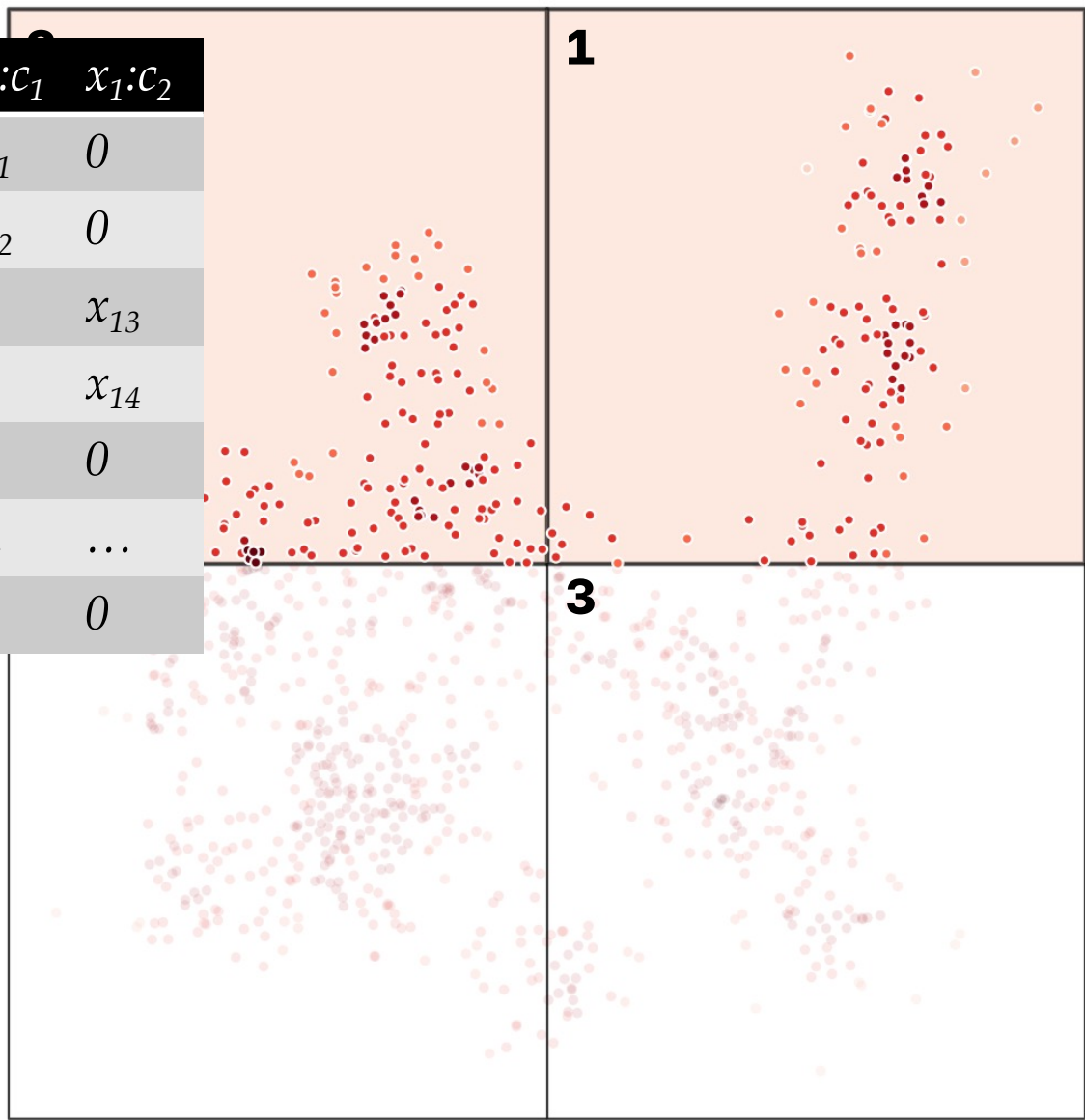


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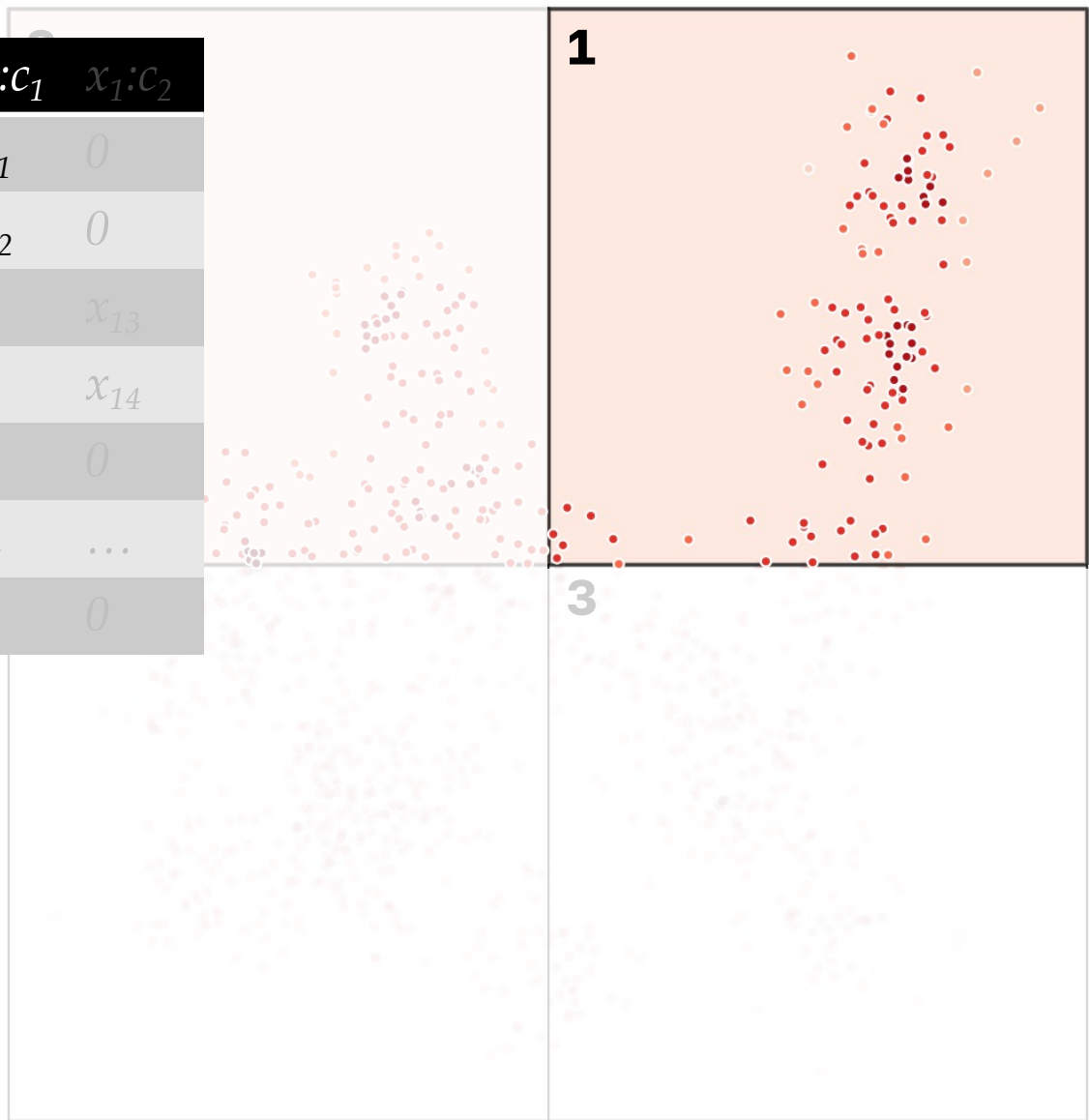
WALK DOWN THE DEPTH LADDER

c	c_1	c_2	$x_1:c$	$x_1:c_1$	$x_1:c_2$
0	1	0	0	x_{11}	0
0	1	0	0	x_{12}	0
0	0	1	0	0	x_{13}
0	0	1	0	0	x_{14}
1	0	0	x_{15}	0	0
...	...	...	...	...	...
1	0	0	x_{1n}	0	0



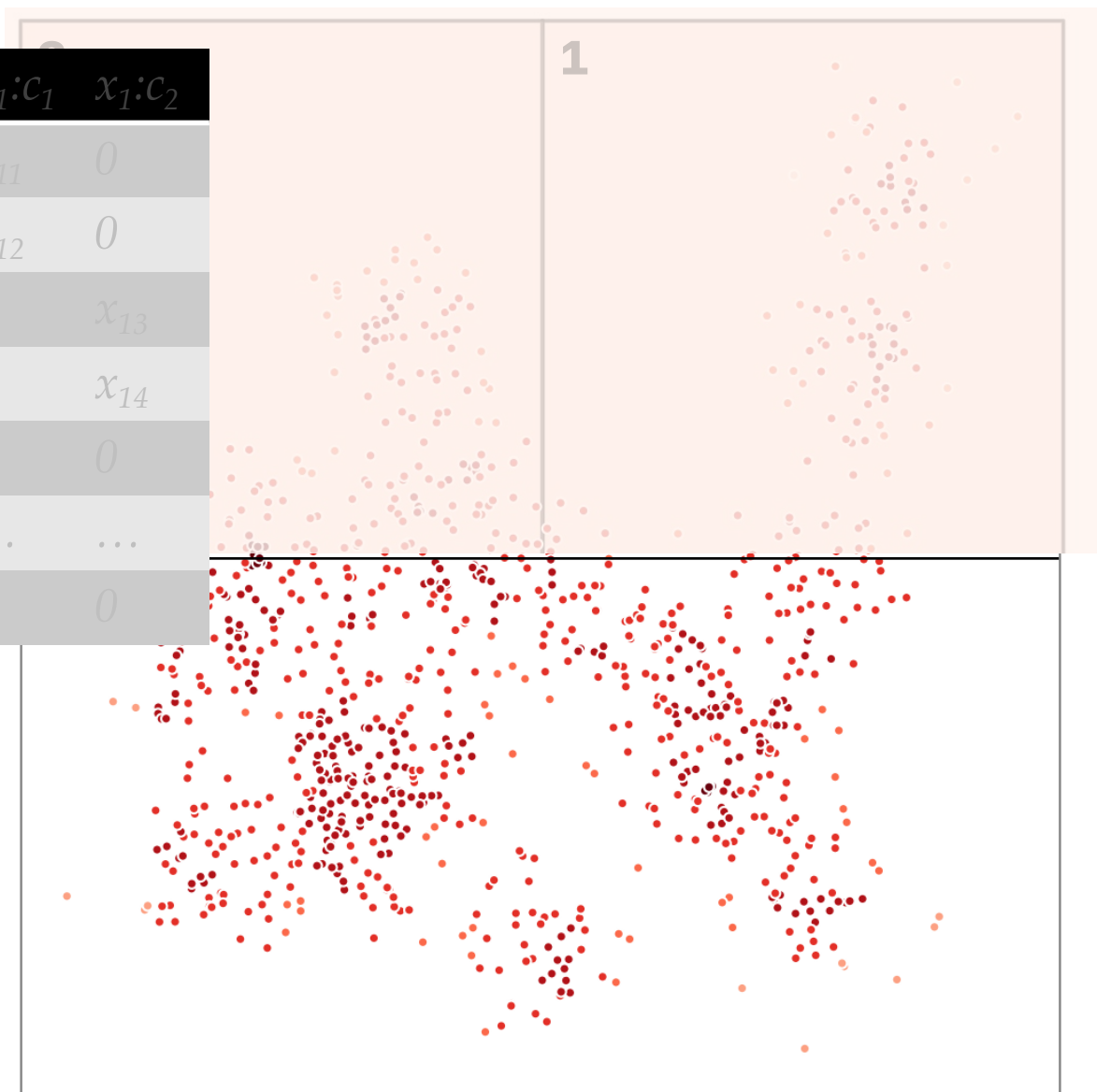
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0	1	0	0	x_{11}	0
0	1	0	0	x_{12}	0
0	0	1	0	0	x_{13}
0	0	1	0	0	x_{14}
1	0	0	x_{15}	0	0
...	...	...	...	...	...
1	0	0	x_{1n}	0	0

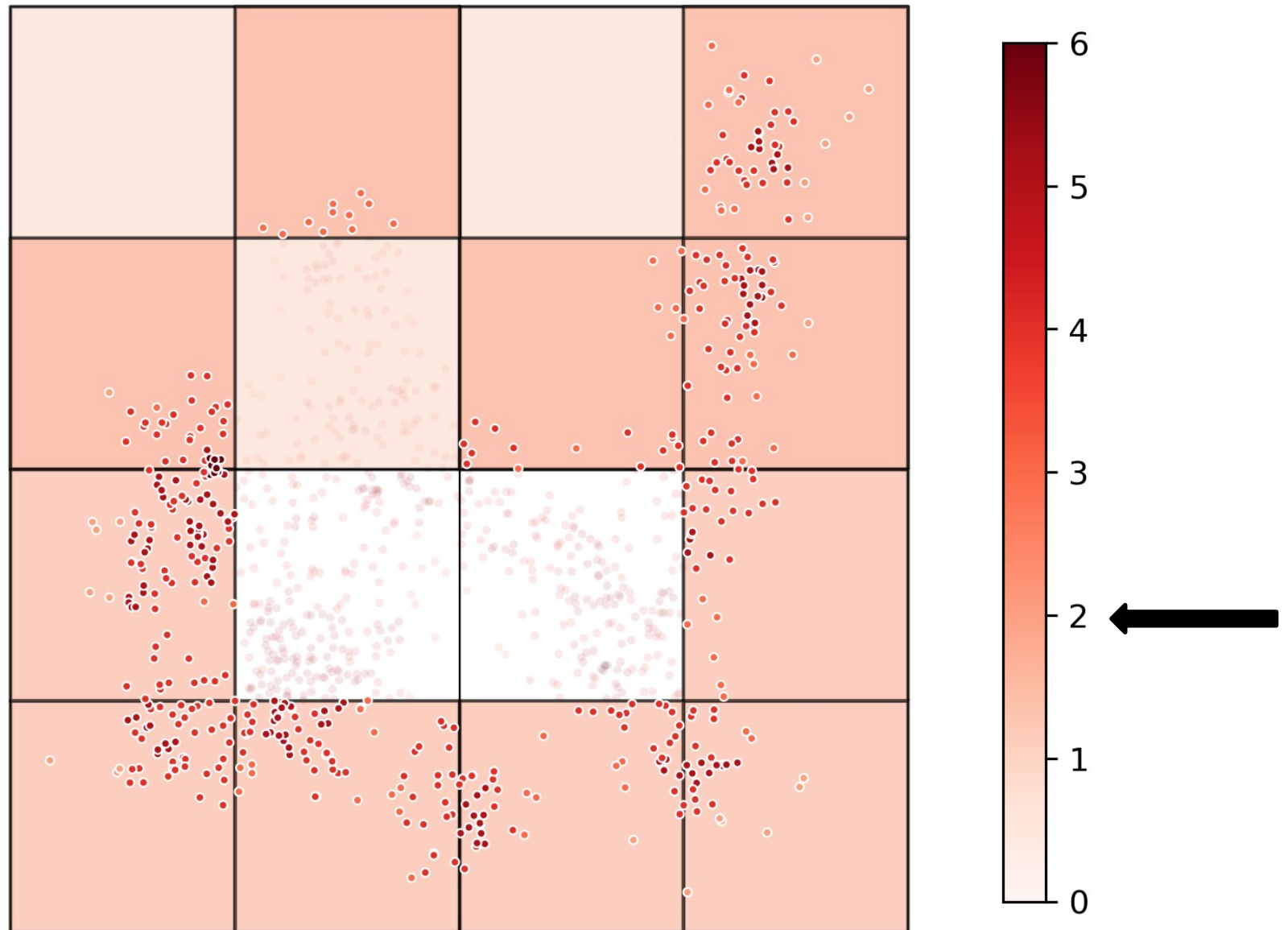


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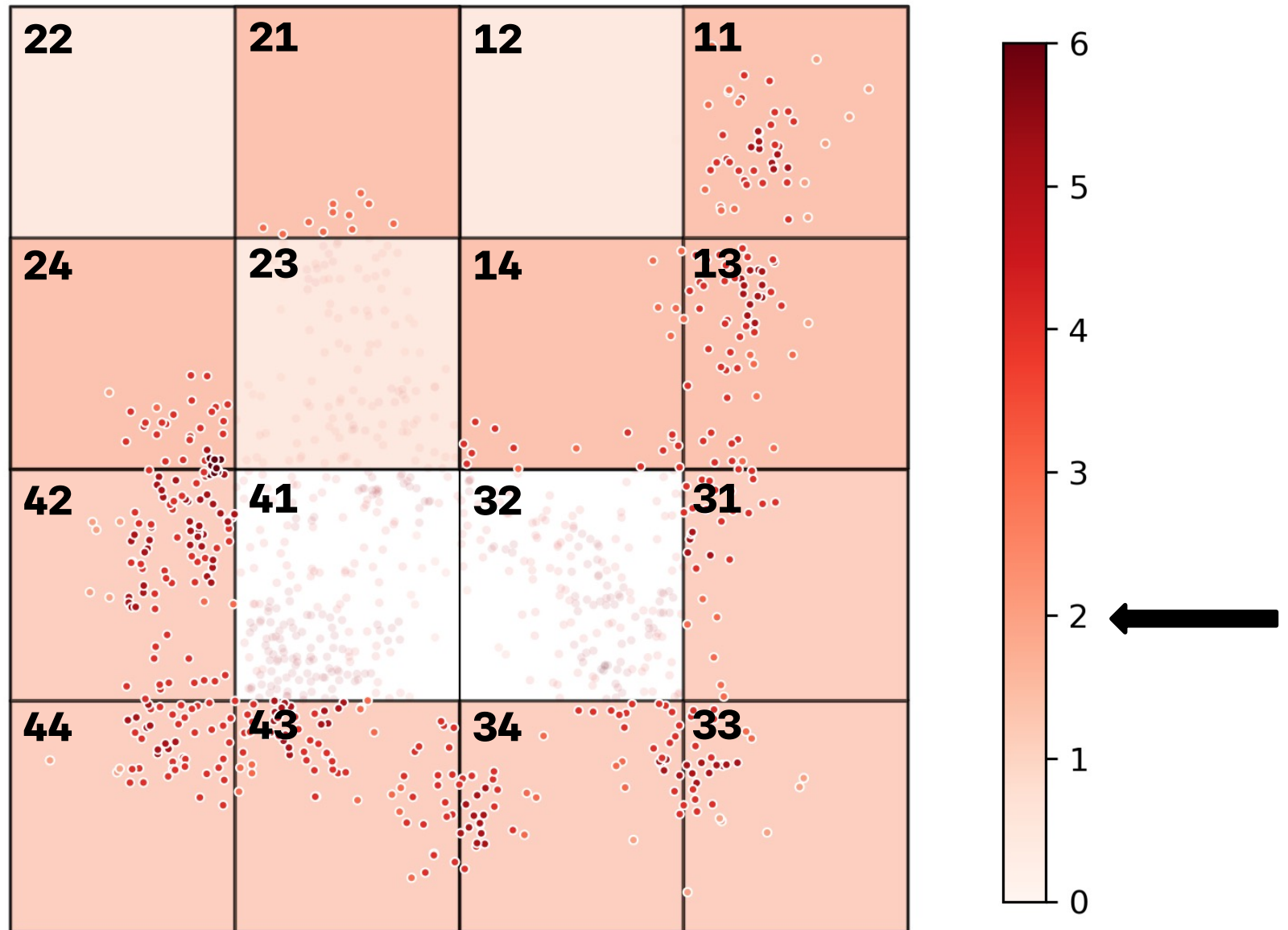
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0	1	0	0	x_{12}	0
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0	0	1	0	0	x_{14}
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...	...	...	...	...	...
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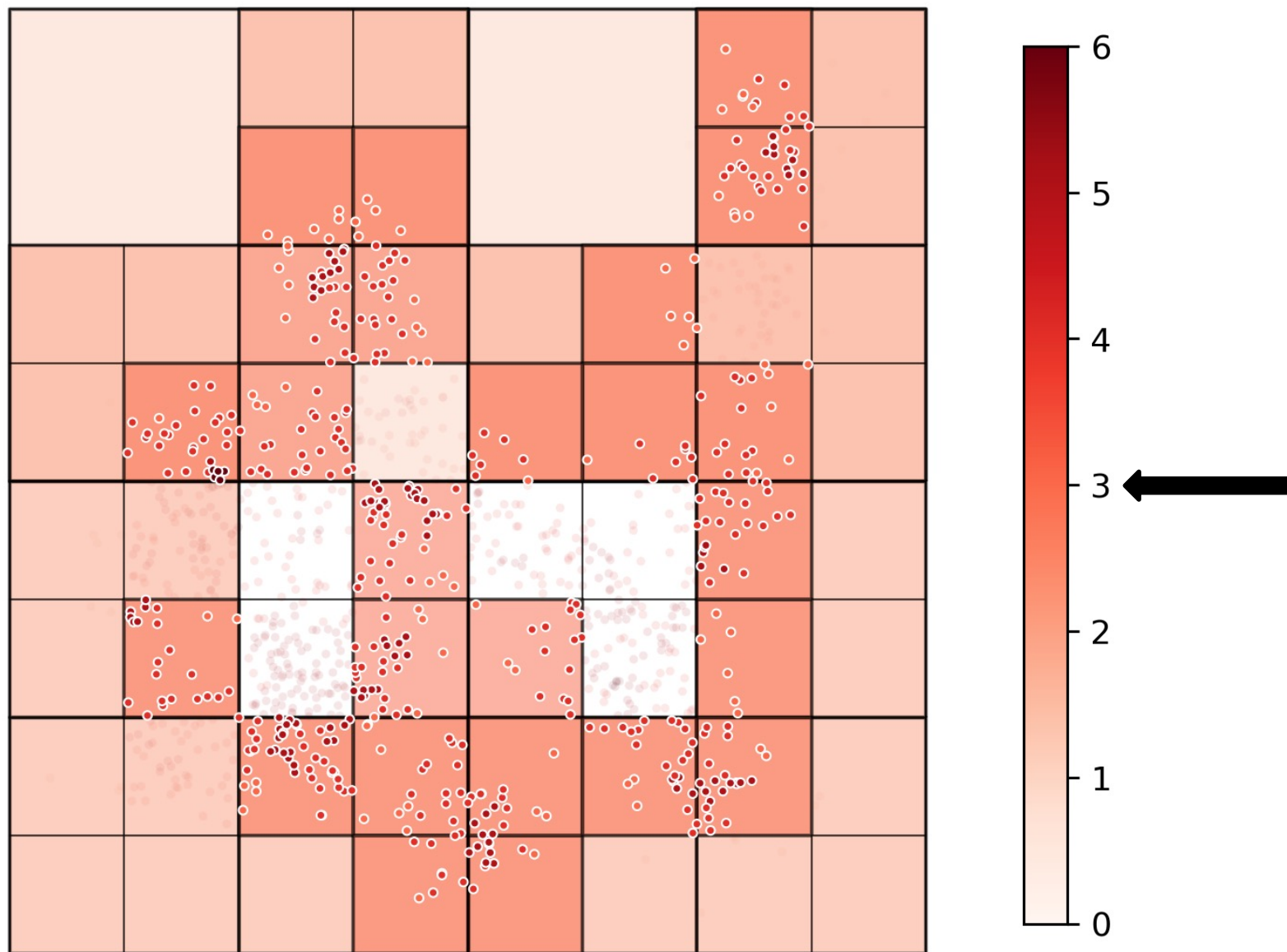
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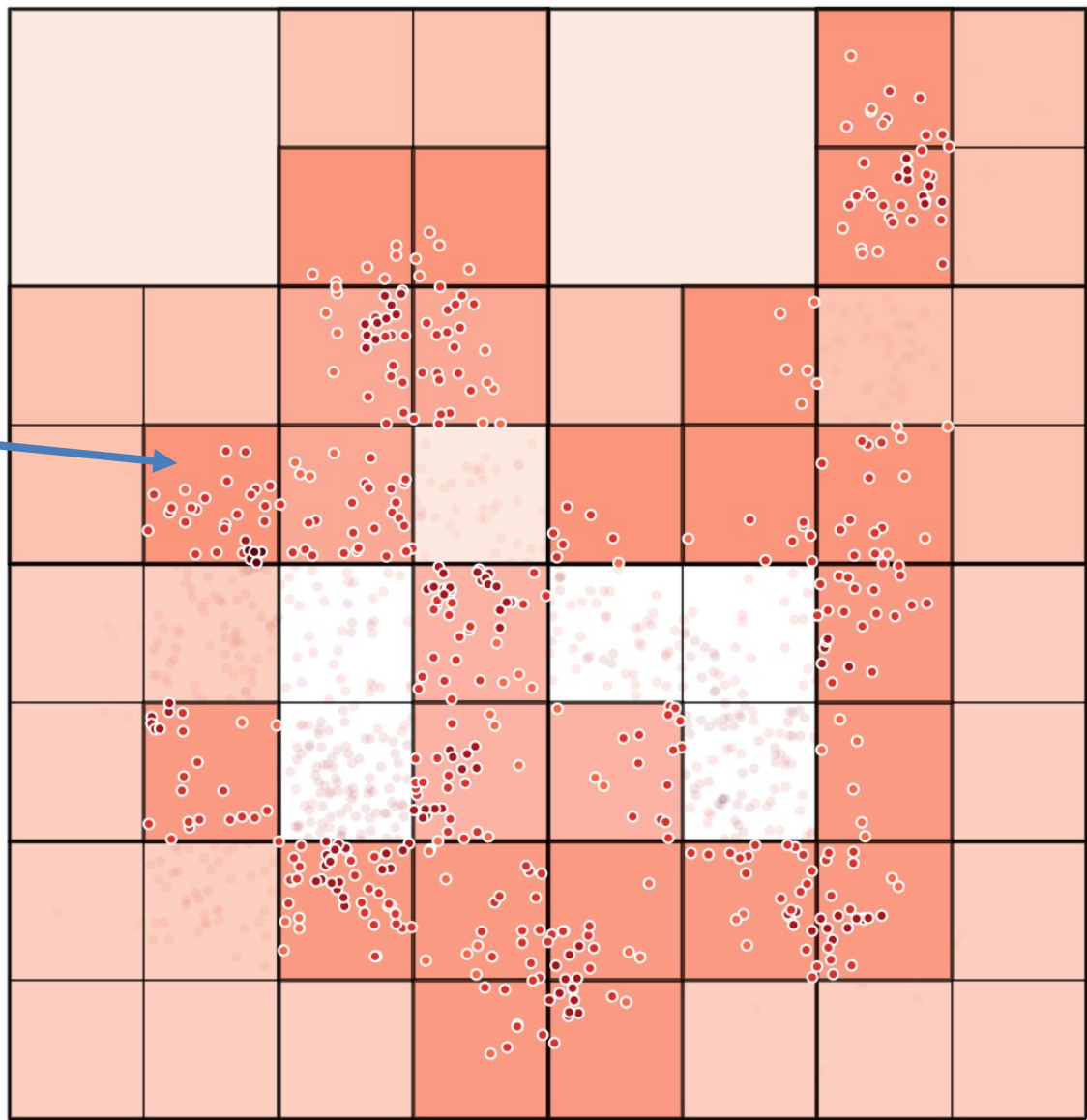


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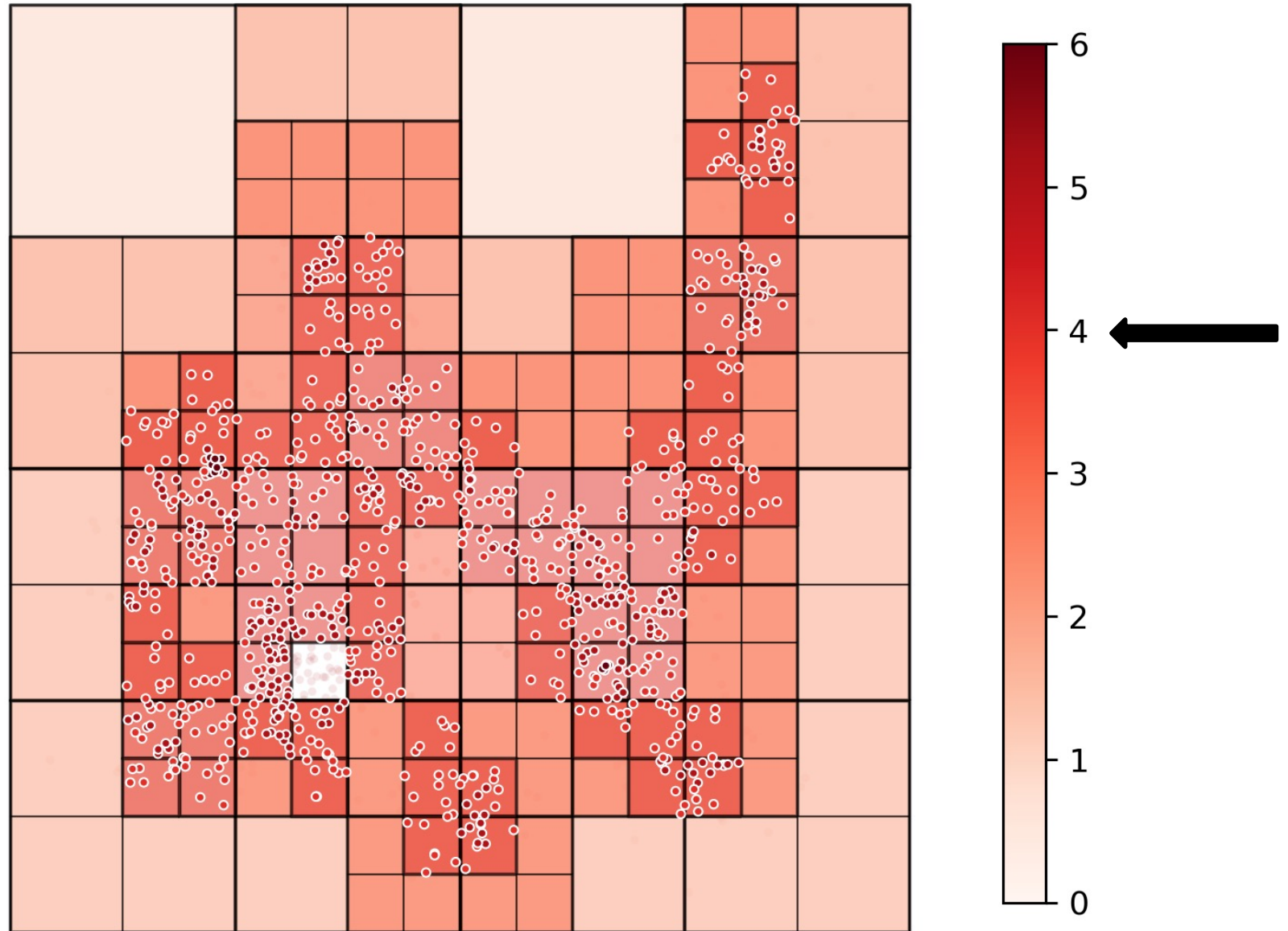


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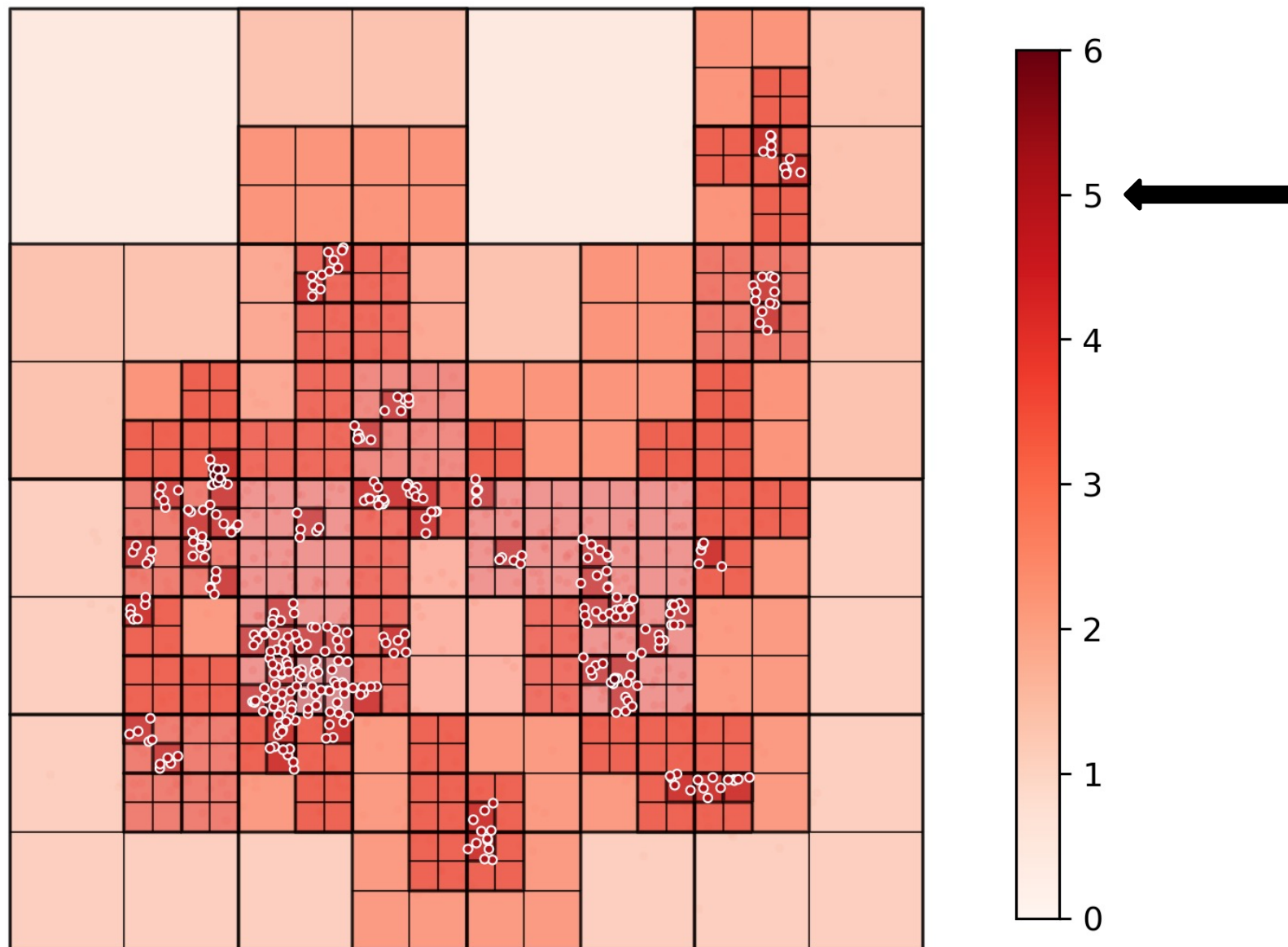
243



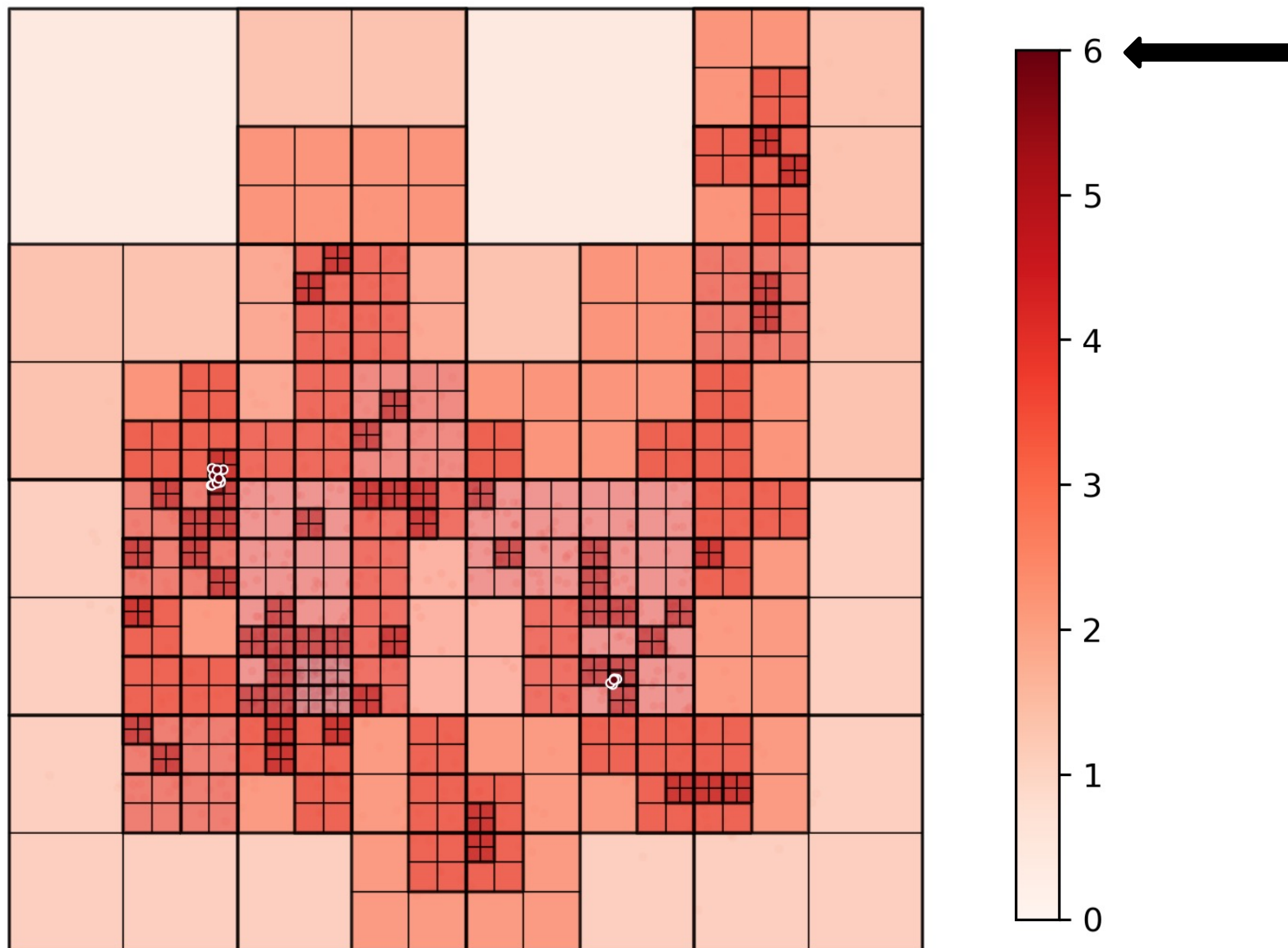
WALK DOWN THE DEPTH LADDER



WALK DOWN THE DEPTH LADDER

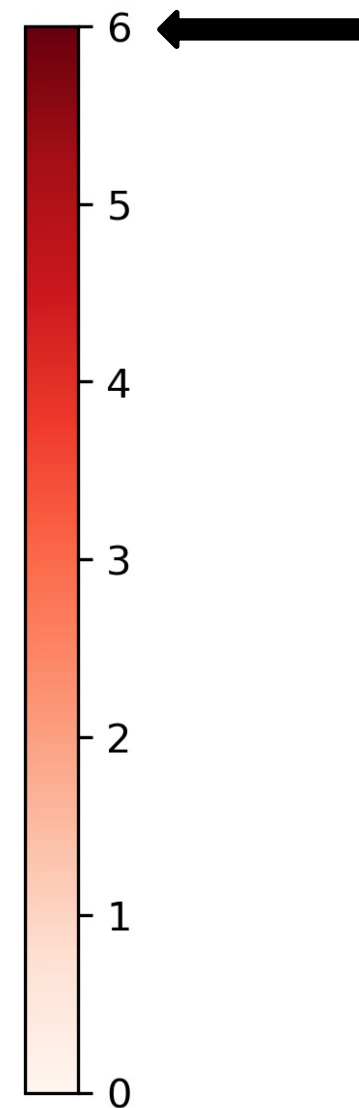
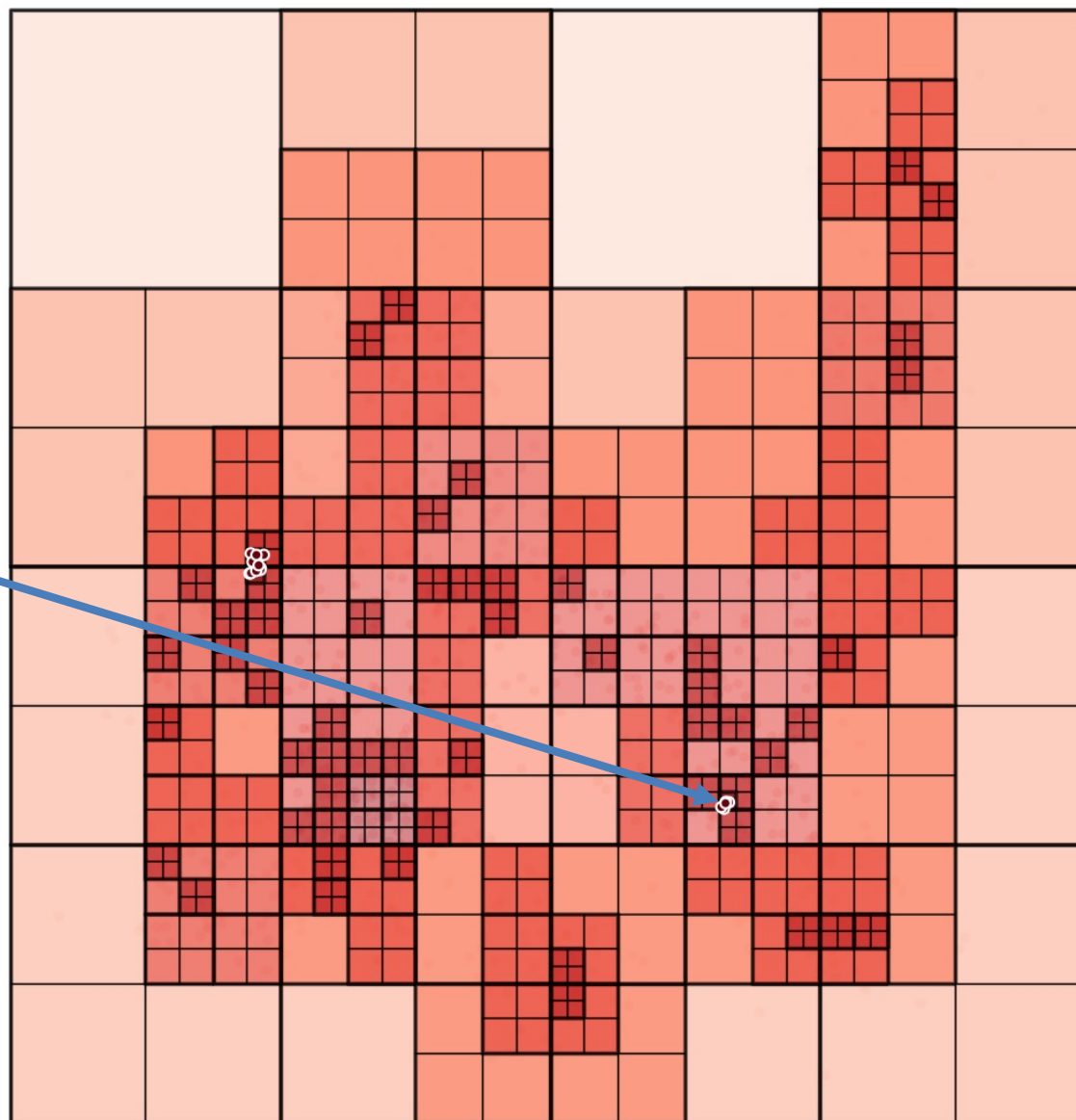


WALK DOWN THE DEPTH LADDER



WALK DOWN THE DEPTH LADDER

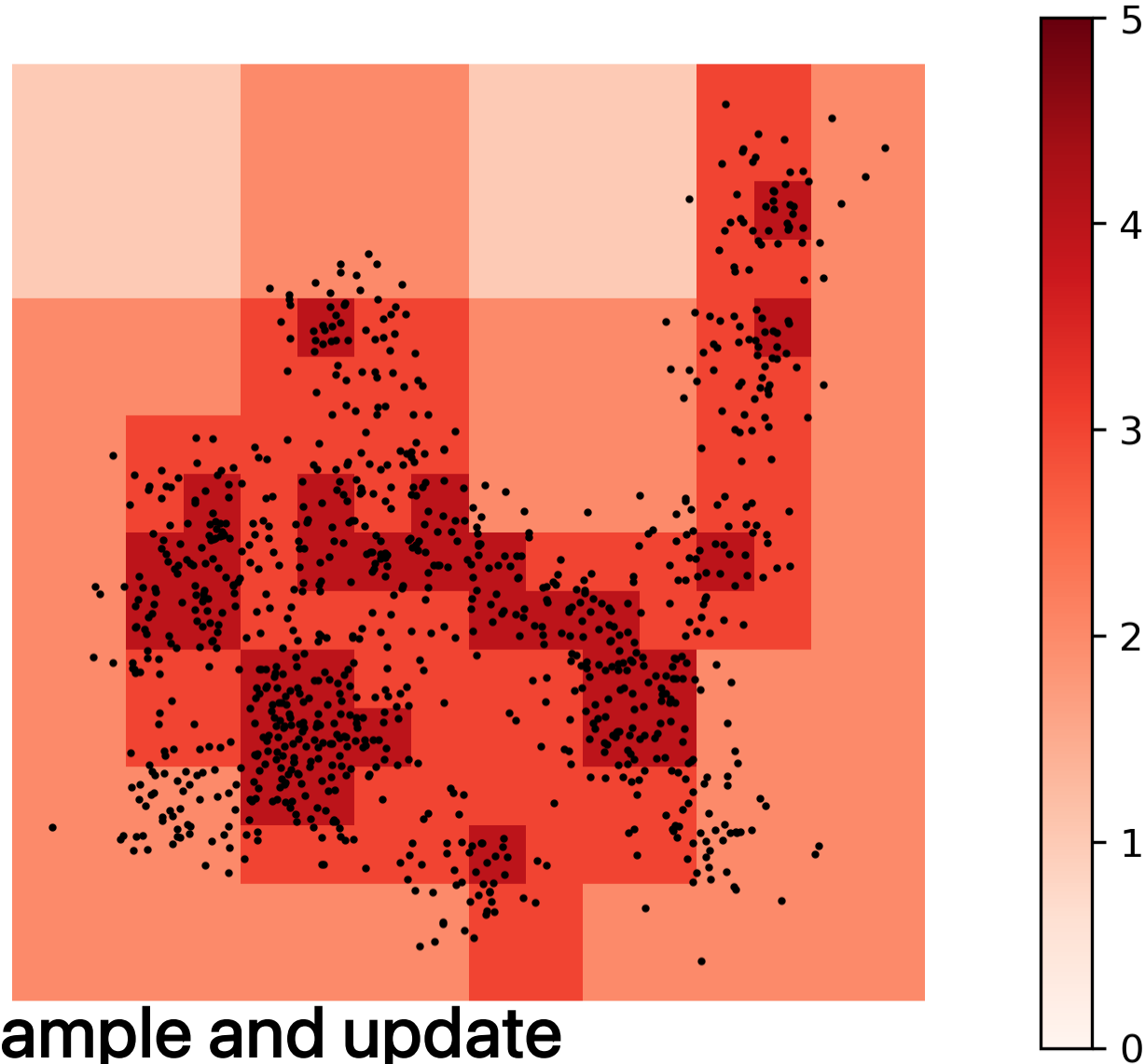
323414



WALK DOWN THE DEPTH LADDER

SuperQT

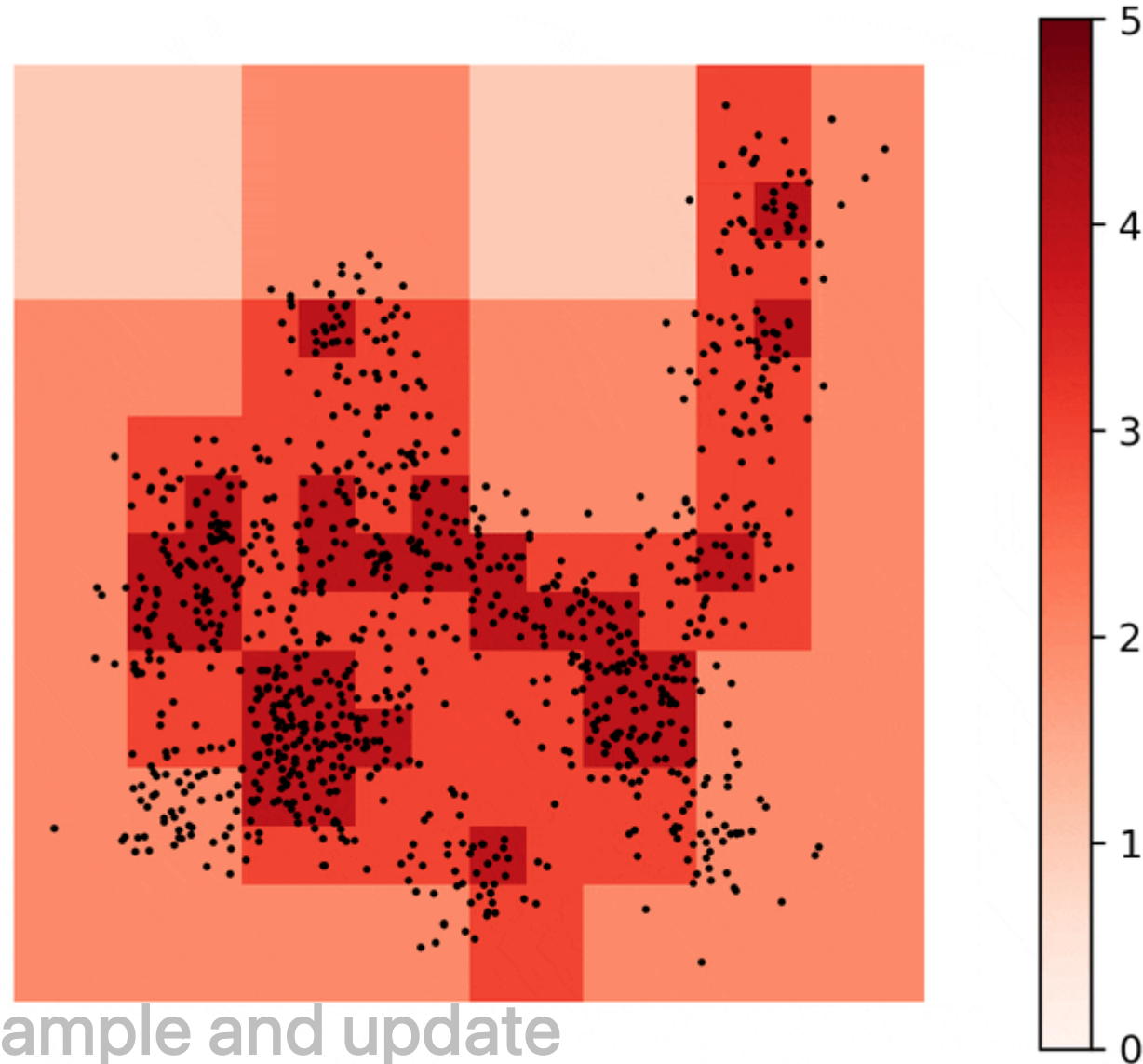
- stable in CV
- totally ordered
- simple out of sample and update



QUADTREES ARE FAIRLY STABLE IN CV

SuperQT

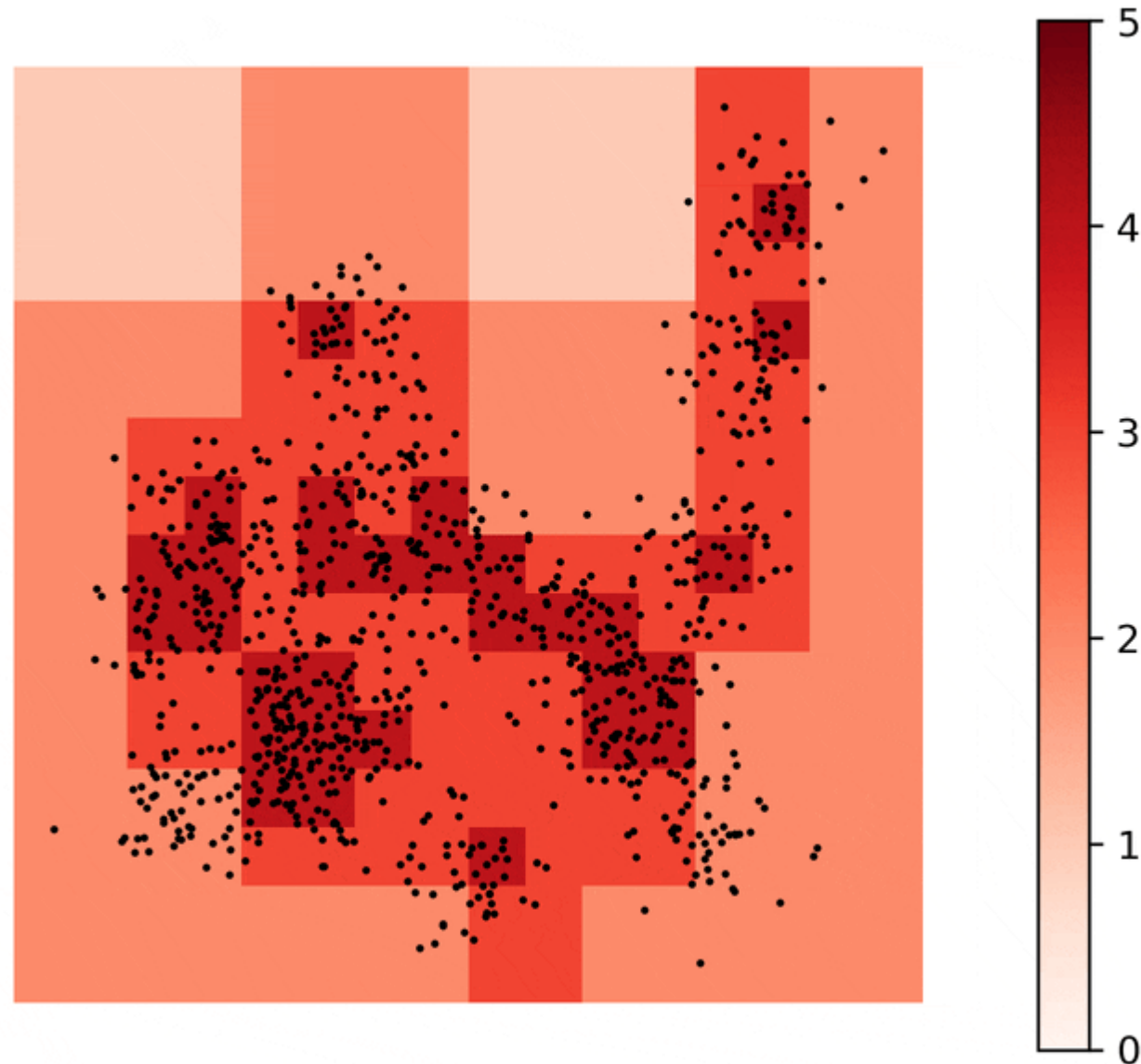
- stable in CV
- totally ordered
- simple out of sample and update



QUADTREES ARE FAIRLY STABLE IN CV

TOTALLY ORDERED

An observation will nest exactly into one cell at any given depth; depth is a function of both fit and density, and big jumps in depth are rare.



QUADTREES ARE FAIRLY STABLE IN CV

OUT-OF-BAG

For interpolating
predictions *within* frame,
we always have a model

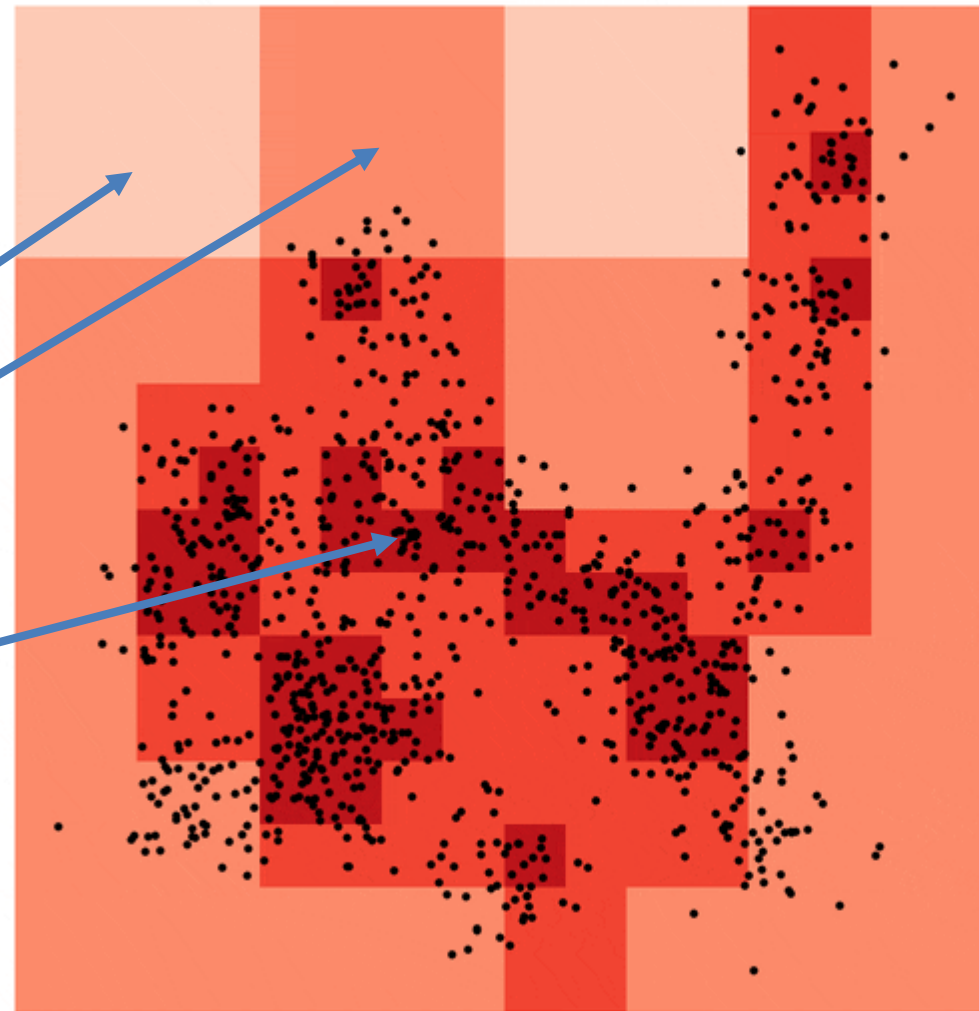
Always **2**

2 or **21**

4112

or

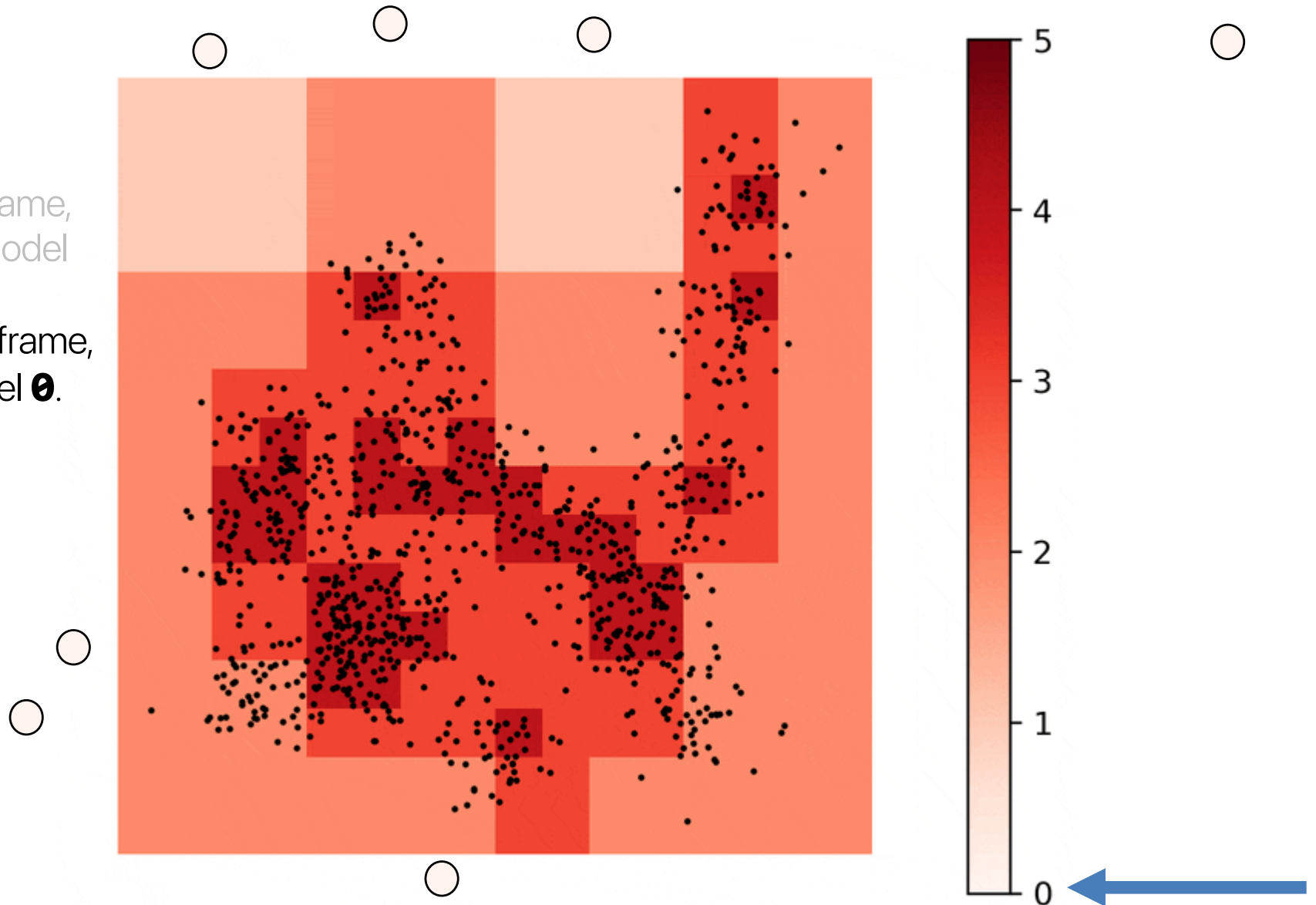
411



QUADTREES ARE FAIRLY STABLE IN CV

OUT-OF-BAG

For interpolating predictions *within* frame, we always have a model
For *extrapolating* predictions outside frame, we always use model **0**.



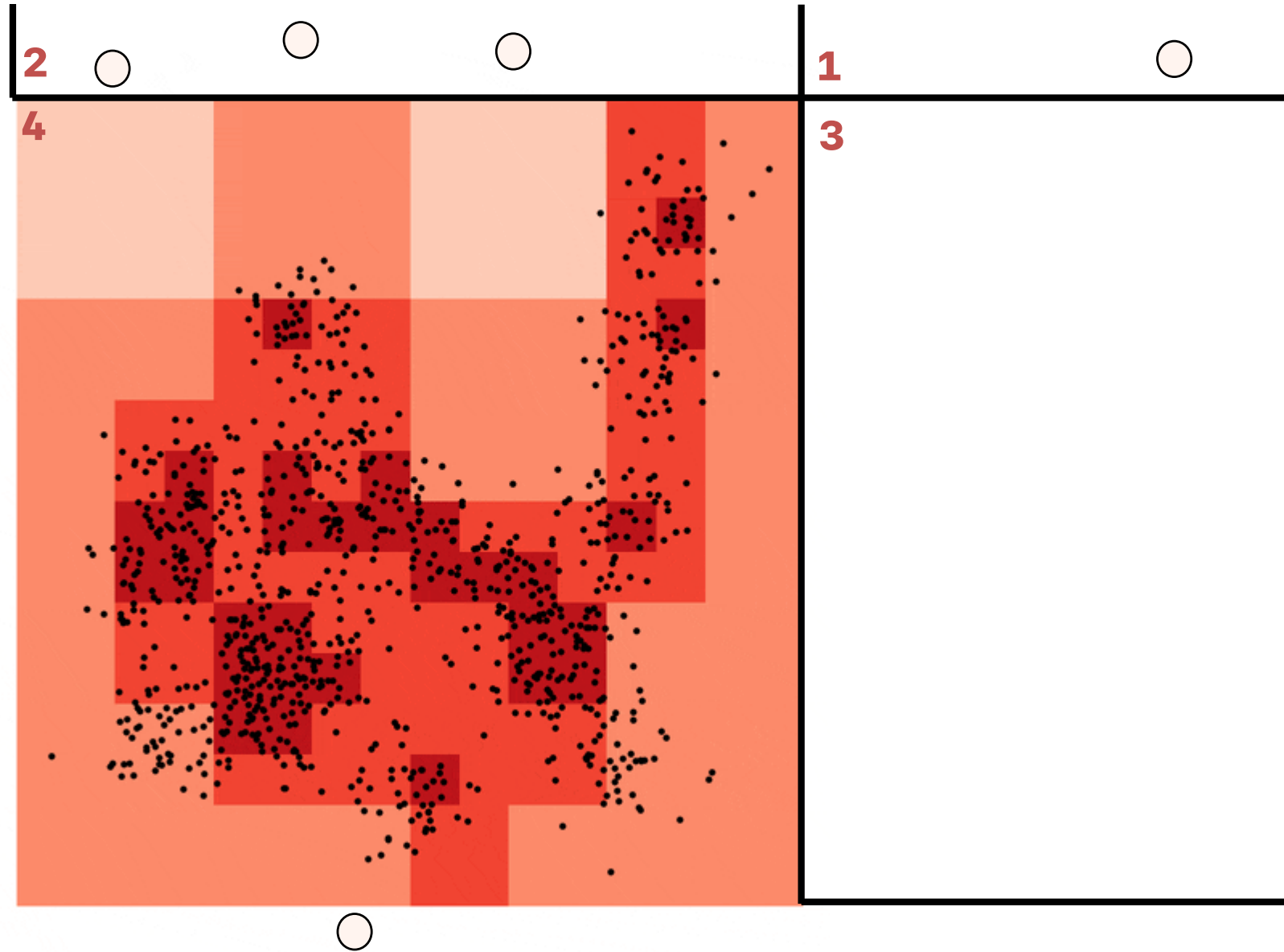
QUADTREES ARE FAIRLY STABLE IN CV

OUT-OF-BAG

For interpolating predictions *within* frame, we always have a model
For *extrapolating* predictions outside frame, we always use model 0.

UPDATING

Treat 0 as a child of some larger frame and prepend all cells with its new index



QUADTREES ARE FAIRLY STABLE IN CV

OUT-OF-BAG

For interpolating predictions *within* frame, we always have a model
For *extrapolating* predictions outside frame, we always use model 0.

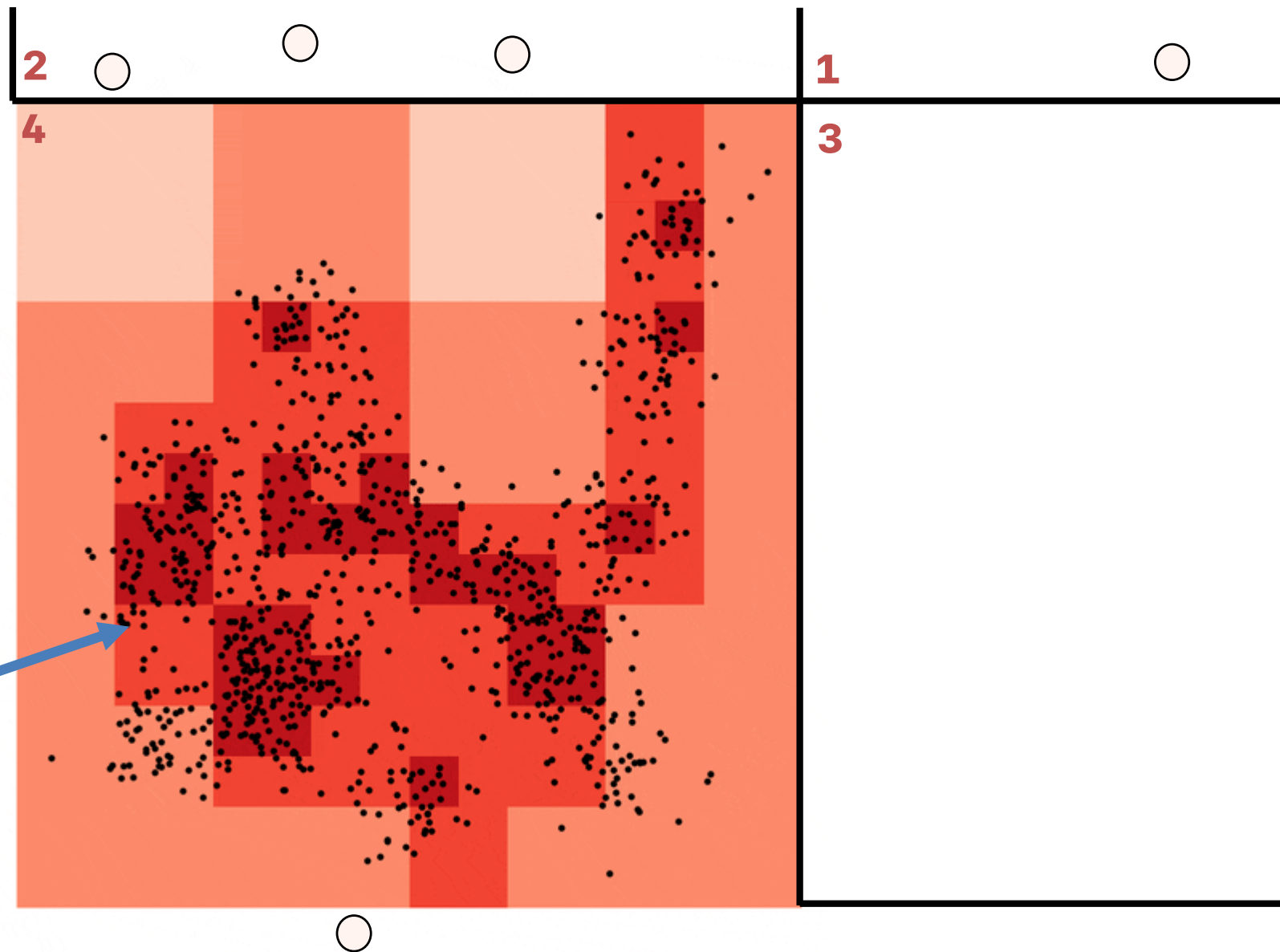
UPDATING

Treat 0 as a child of some larger frame and prepend all cells with its new index

424

updates to

4424



QUADTREES ARE FAIRLY STABLE IN CV

0. INITIALIZE: add root node to queue with label 0 and global model predicting $y \sim X$

1. SPLIT: pop the first node from our queue and split. For each child in that split:

1-1. fit a submodel predicting only child's outcomes using child's data.

1-2. update predictions for the whole map.

1-3. if child model is significantly more skillful than parent **within child (out of sample)**, label samples with child label, return child to queue, & go to 1; otherwise assign all samples to parent's label & drop child.

2. PRUNE: fit a model with all current feature:label interactions (one hot).

2-1. if all terms are helpful, go to 3,

otherwise find the depth (d^*) of the least helpful feature:label term

2-2. for each unhelpful term at d^* , merge the feature:child_label term into feature:parent_label, and return to 2.

3. FINALIZE: Fit a final model on final retained feature:label interaction terms

THE SUPERVISED QUADTREE

0. INITIALIZE: add root node to queue with label 0 and global model predicting $y \sim X$

1. SPLIT: pop the first node from our queue and split. For each child in that split:

1-1. fit a submodel predicting only child's outcomes using child's data.

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3. FINALIZE: Fit a final model on final retained feature:label interaction terms

Note that this is *model agnostic*: can use any learner, classification or regression.

- GLMs can split using LRT (χ^2) prune using Wald (t) tests, placing a natural limit on n_{child}

- Always can split using score improvement ($\Delta MSE > \epsilon$)

prune using permutation feature importance ($X_p * I(leaf)$ in best $q\%$ of unpruned features)

THE SUPERVISED QUADTREE

0. INITIALIZE: add root node to queue with label 0 and global model predicting $y \sim X$

1. SPLIT: pop the first node from our queue and split. For each child in that split:

1-1. fit a submodel predicting only child's outcomes using child's data.

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2-2. for each unhelpful term at d^* , merge the feature:child_label term into feature:parent_label, and return to 2.

3. FINALIZE: Fit a final model on final retained feature:label interaction terms

Note that this is *tree agnostic*: can use any spatial tree index

- Pruning is most useful with hierarchical trees, where parent is still "local"
- Overlap (e.g. in RTree) introduces intersection terms (feature:label1:label2).
- Balance is optional, but helpful for submodel efficiency

THE SUPER* TREE

CLUSTERING REGRESSION

jointly solving clustering and regression

GEOGRAPHICAL CLUSTER-REG

jointly solving clustering and regression

UNDERSTANDING QUADTREES

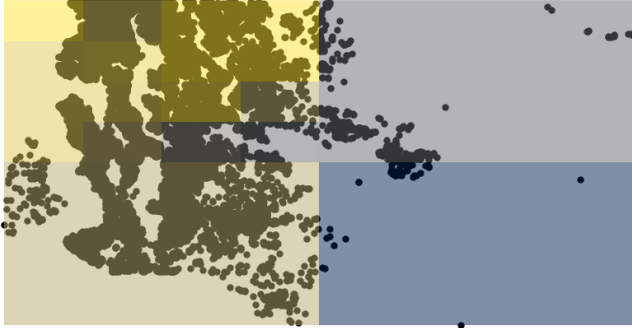
spatial splits for spatial fits

APPLYING QUADTREE REGRESSION

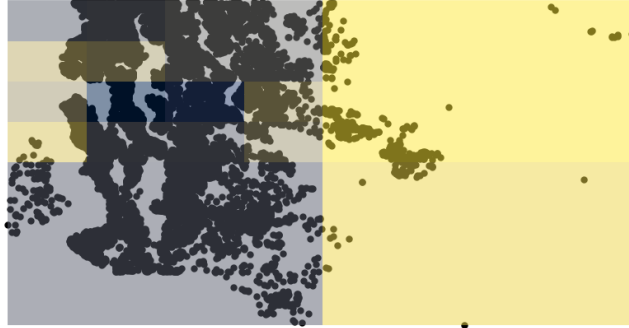
model comparison & basic metrics

THE SUPERVISED QUADTREE

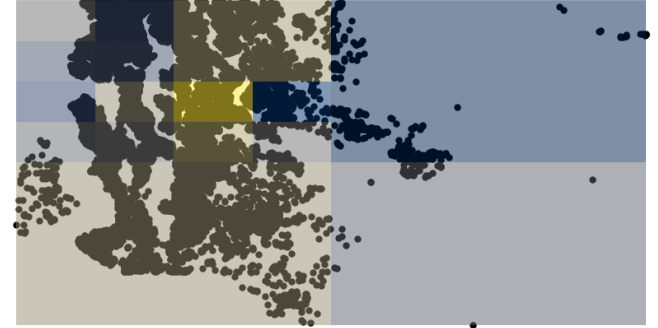
intercept



bedrooms



bathrooms



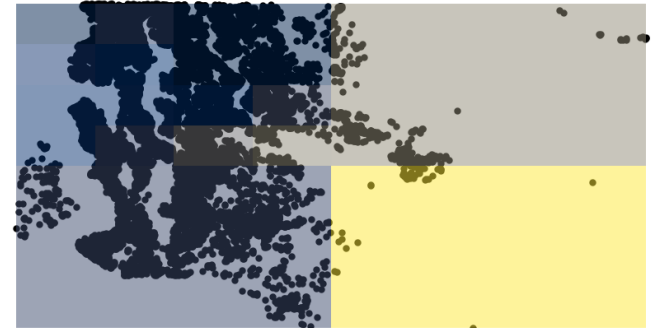
sqft_living



sqft_lot

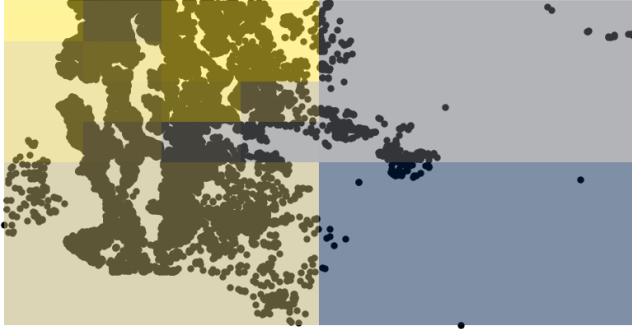


yr_built



COEFFICIENT SURFACES LOOK LIKE THIS

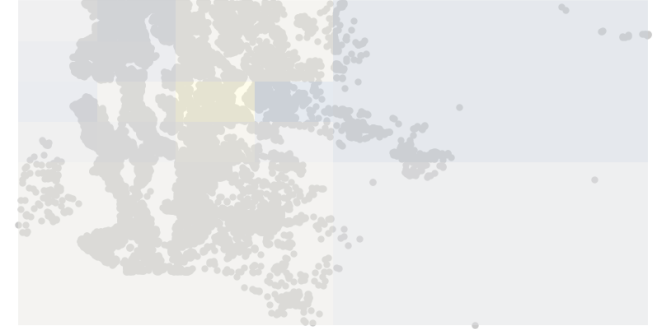
intercept



bedrooms



bathrooms



sqft_living



sqft_lot



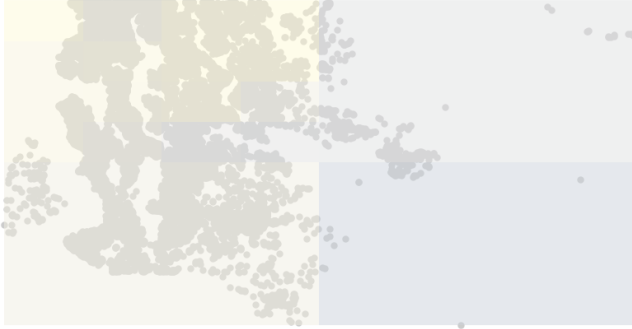
yr_built



Prices in general are expensive towards the coast, also in the techy areas in the northwest.

COEFFICIENT SURFACES LOOK LIKE THIS

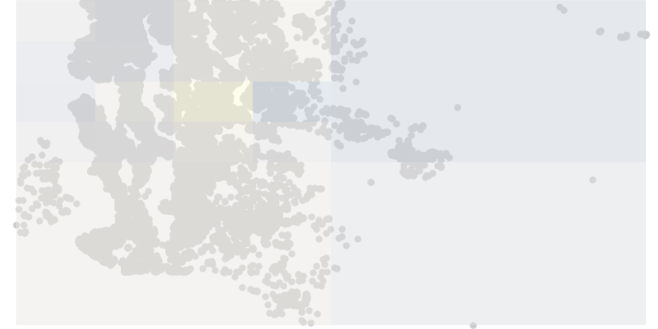
intercept



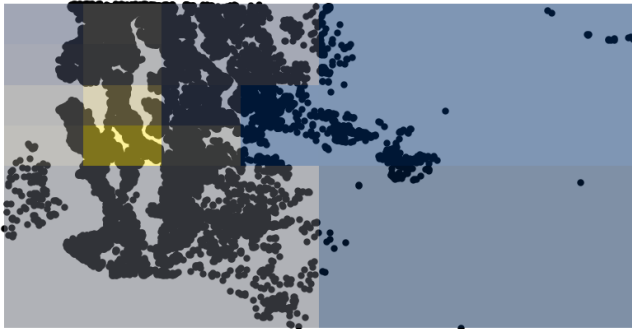
bedrooms



bathrooms



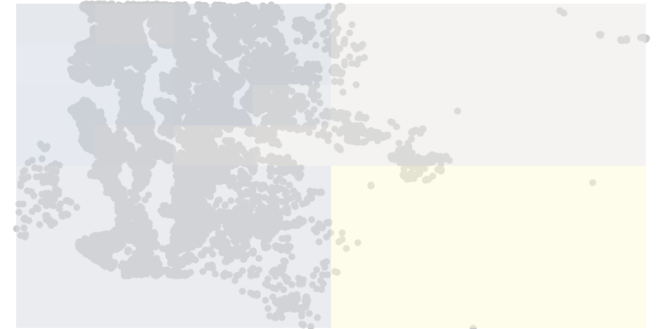
sqft_living



sqft_lot



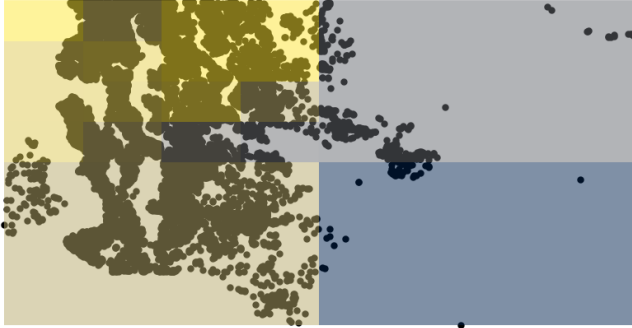
yr_built



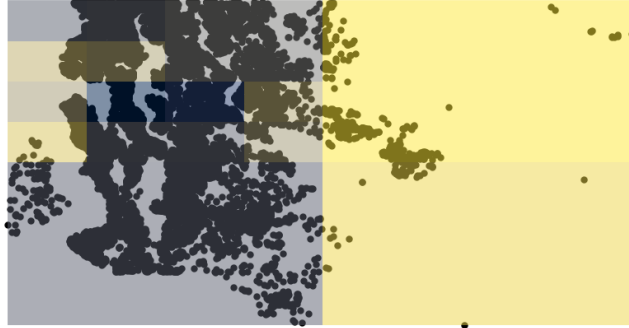
Space is at a premium, with living space slightly more expensive south-center.

COEFFICIENT SURFACES LOOK LIKE THIS

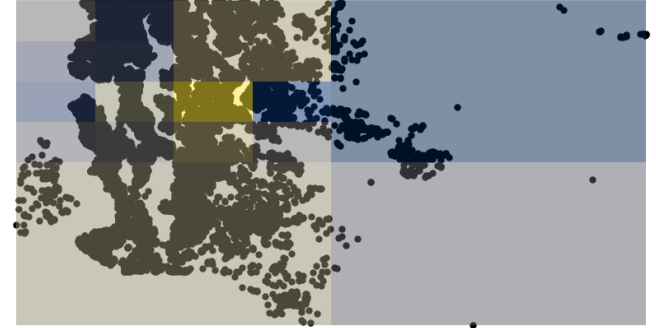
intercept



bedrooms



bathrooms



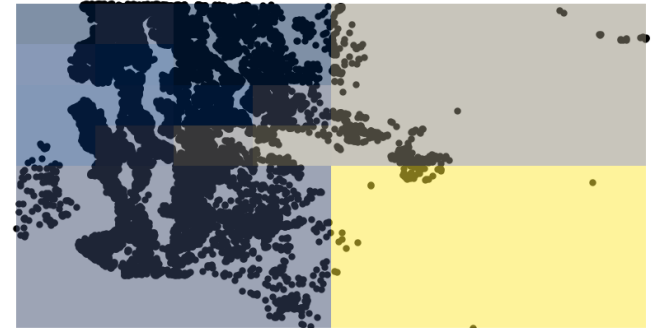
sqft_living



sqft_lot



yr_built



COEFFICIENT SURFACES LOOK LIKE THIS

PLACE	METHOD	MSE	R ²	TIME
Baltimore	OLS	.16	.49	Sub-seconds
<i>n</i> =211	Skater 15 Regimes	.02	.93	Seconds
	MGWR	.01	.95	Seconds
	SuperQT (no prune)	.04 (.02)	.89 (.94)	Seconds
King's Co.	OLS	.13	.53	Seconds
<i>n</i> =31k	Skater 15 Regimes	.05	.811	Hours
	MGWR	DNF	DNF	DNF
	SuperQT (no prune)	.06 (.06)	.78 (.78)	Minutes

COMPARISONS

PLACE	METHOD	MSE	R ²	TIME
Baltimore	OLS	.16	.49	Sub-seconds
<i>n</i> =211	Skater 15 Regimes	.02	.93	Seconds
	MGWR	.01	.95	Seconds
	SuperQT (no prune)	.04 (.02)	.89 (.94)	Seconds
King's Co.	OLS	.13	.53	Seconds
<i>n</i> =31k	Skater 15 Regimes	.05	.811	Hours
	MGWR	DNF	DNF	DNF
	SuperQT (no prune)	.06 (.06)	.78 (.78)	Minutes

Small n: usually need eps+perm (lrt+wald dof issues)

Big n: both lrt+wald and eps+perm are useful

COMPARISONS

PLACE	METHOD	MSE	R ²	TIME
Baltimore	OLS	.16	.49	Sub-seconds
<i>n</i> =211	Skater 15 Regimes	.02	.93	Seconds
	MGWR	.01	.95	Seconds
	SuperQT (no prune)	.04 (.02)	.89 (.94)	Seconds
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COMPETITIVE EVEN AFTER PRUNING

PLACE	METHOD	MSE	R ²	TIME
Baltimore	OLS	.16	.49	Seconds
<i>n</i> =211	Skater 15 Regimes	.02	.93	Seconds
	SuperQT (no prune)	.04 (.02)	.89 (.94)	Seconds
King's Co.	OLS	.13	.53	Seconds
<i>n</i> =31k	Skater 15 Regimes	.05	.811	Hours
	MGWR	DNF	DNF	DNF
	SuperQT (no prune)	.06 (.06)	.78 (.78)	Minutes

Pruning is especially powerful in big data:

~90 feature x leaf interactions in Skater & SuperQT

~60 post pruning w/ very small change to fit (for both)

COMPETITIVE EVEN AFTER PRUNING

MIT-Licensed

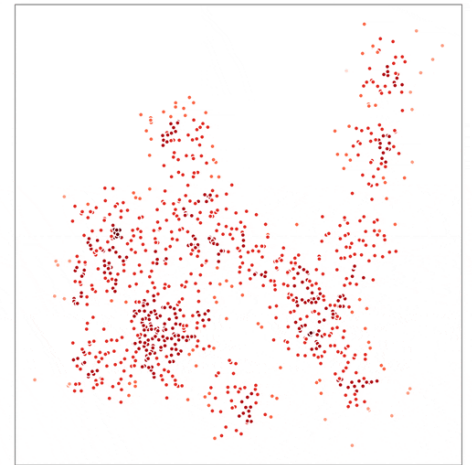
High performance w/ `scipy.sparse`

Available in PySAL by December 2025

THE SUPERVISED QUADTREE, IMPLEMENTED

SUPER * TREE LEARNERS:

a new metalearner for
local data science



LEVI JOHN WOLF `levi.john.wolf@bristol.ac.uk`